

Primeros pasos en IoT con LoRaWAN

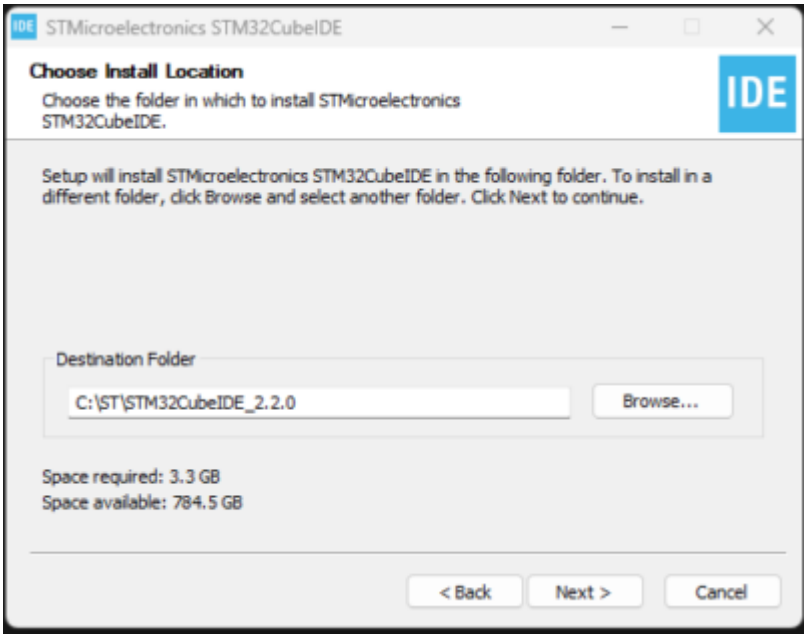
Entorno de desarrollo

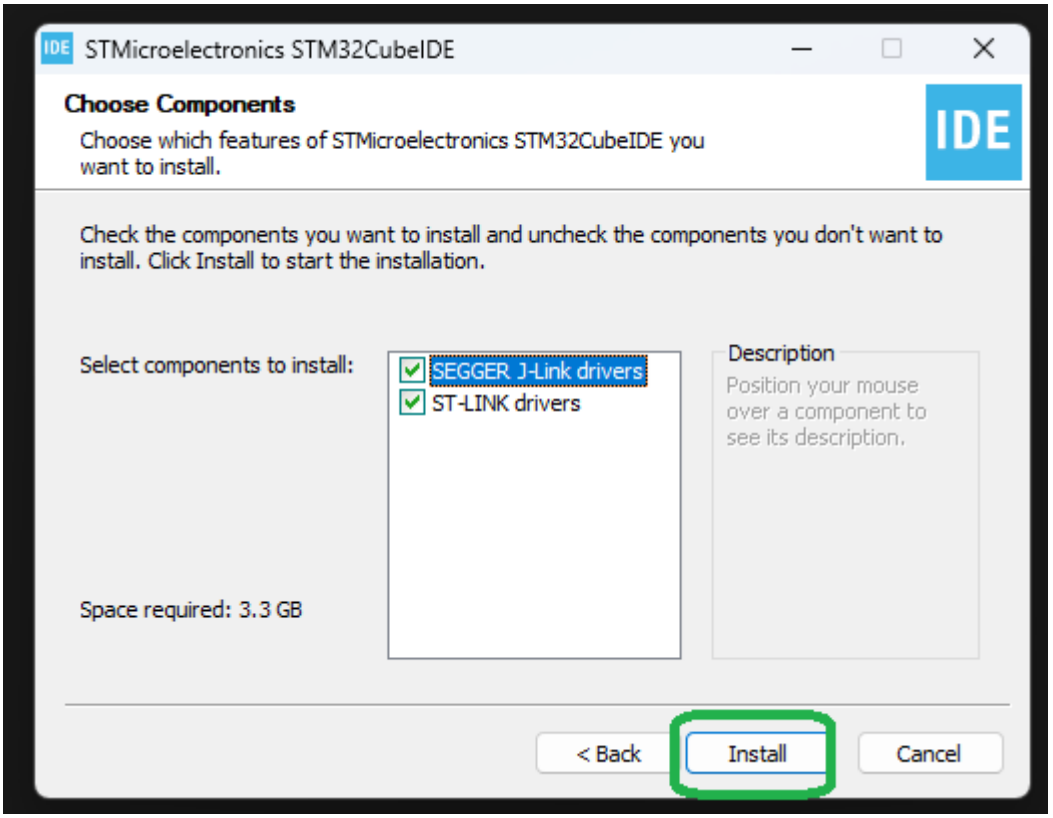
- Hardware/Firmware
  - Puerta de enlace(opcional): NUCLEO-F746ZG + LRWAN GS HF1 LoRa
  - Placa de desarrollo: STM32WL55JC1, Firmware STM32WL55JCIx: V1.3.1 + Controladores
  - Placa de circuito impreso Apiariot: Disponible próximamente.
- *Software:*
  - STMCubeIDE(V:2.2.0)
  - STM32CubeMX(V:6.18.1)
  - STMCubeWL(V1.6.0/1): Biblioteca para STM32WL55JC
  - Firmware de la placa de desarrollo STM32WL55JCIx, V1.3.

Preparación del entorno (conjunto de herramientas)

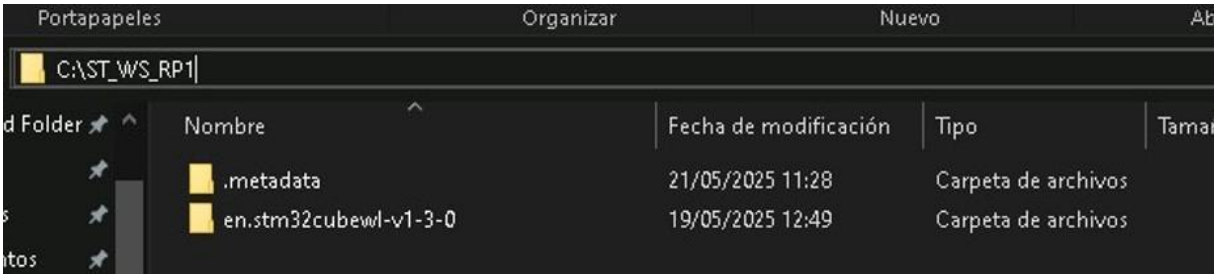
Se recomienda mantener la misma ruta de carpeta y seguir cada paso al pie de la letra.

- Descarga e instala el entorno de desarrollo integrado (IDE) ”STM32CubeIDE”.

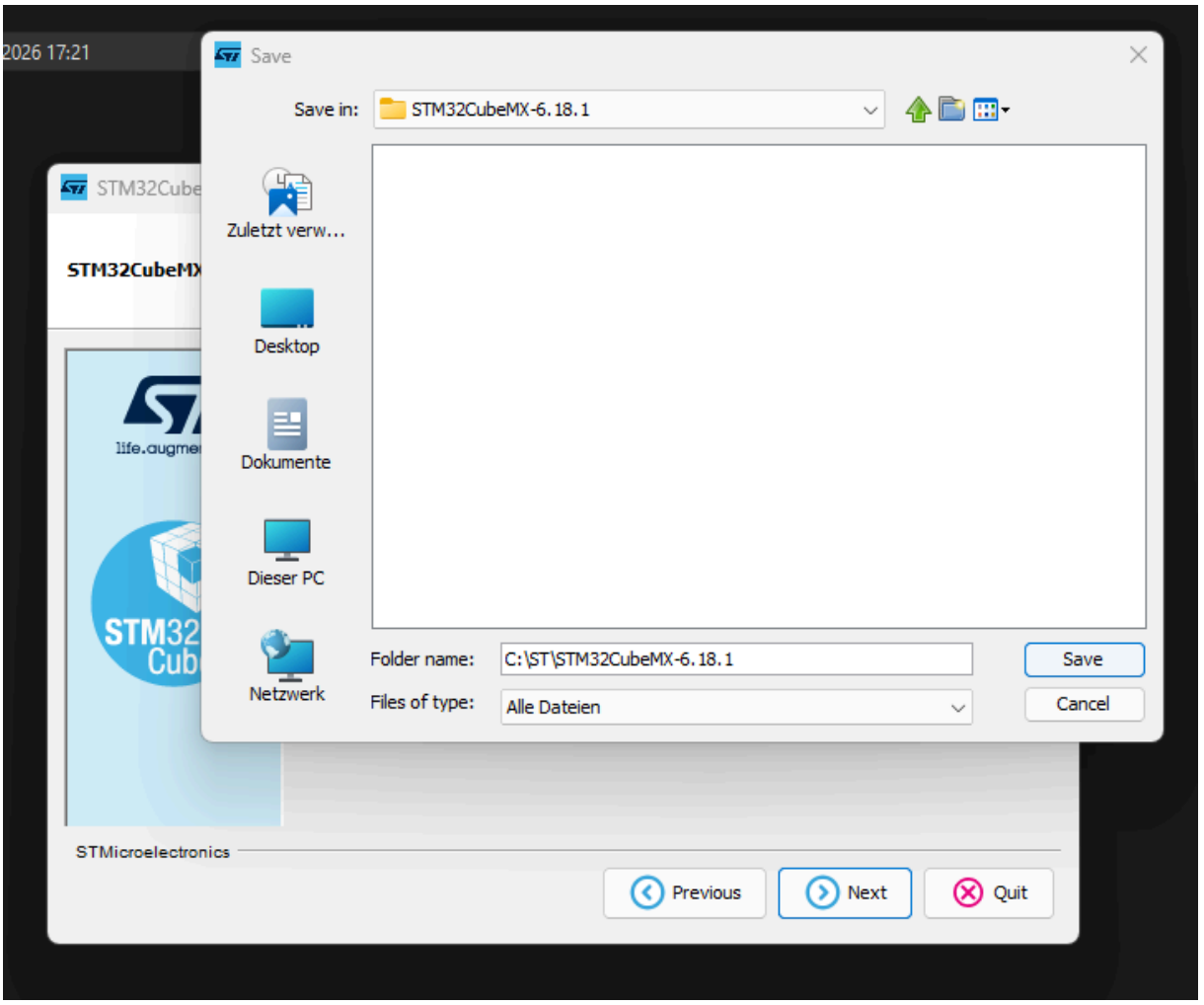


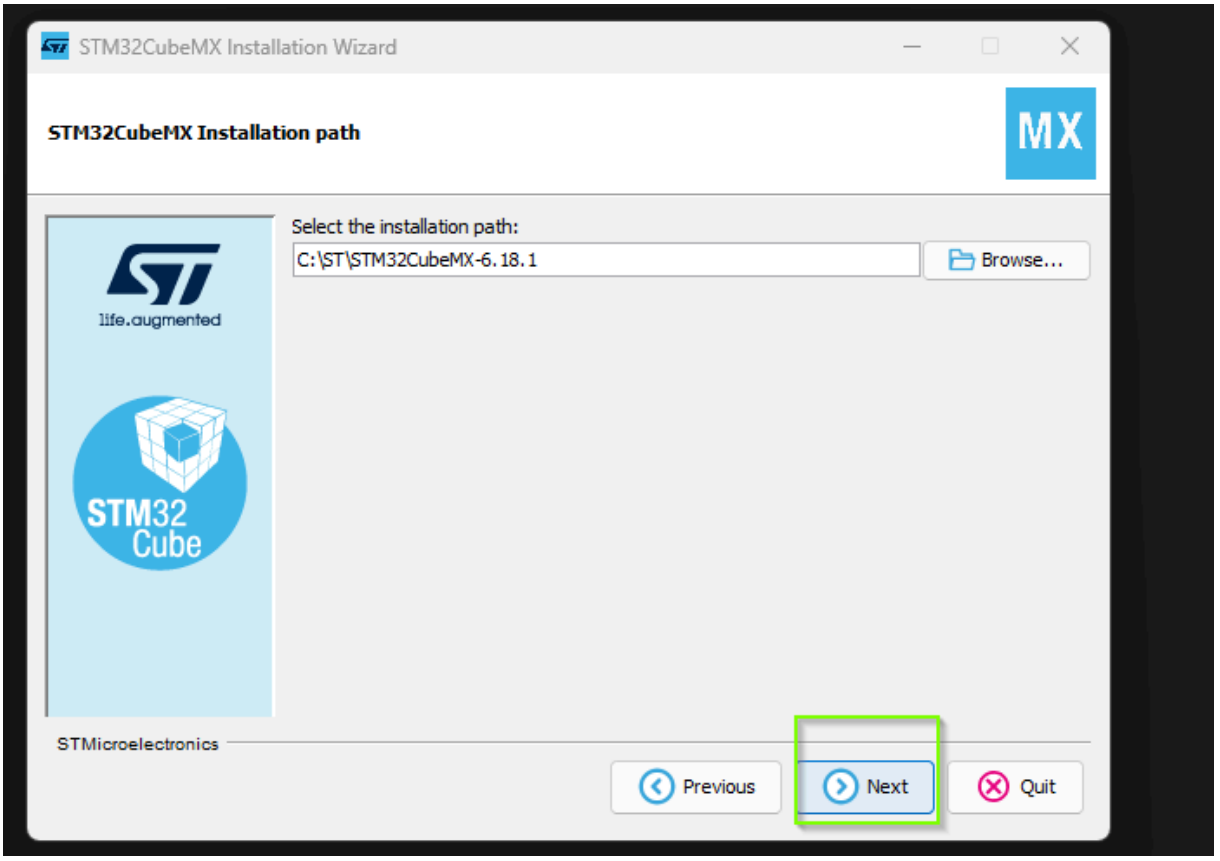


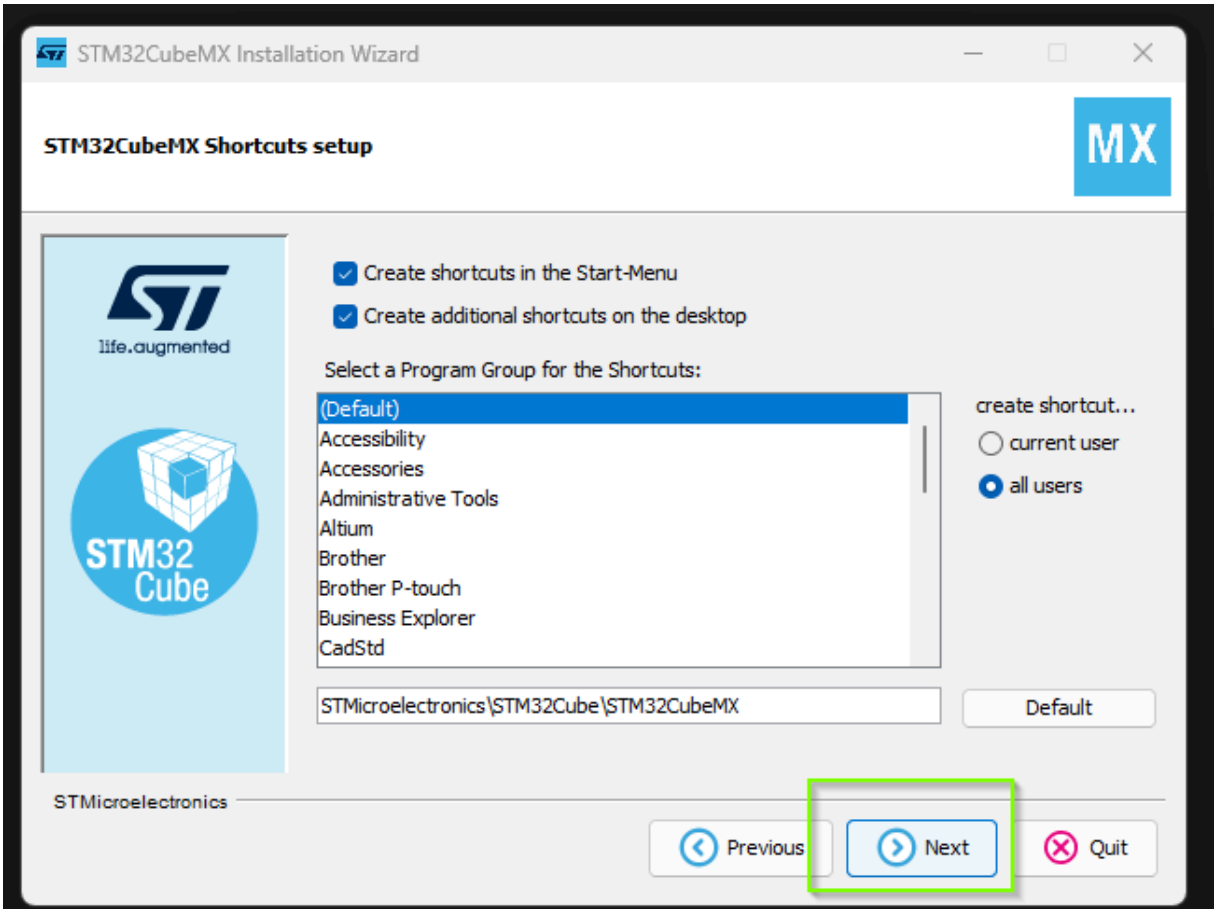
- Crea la carpeta "C:\ST_WS_RP1".
  Se recomienda utilizar la misma carpeta y la misma ruta.



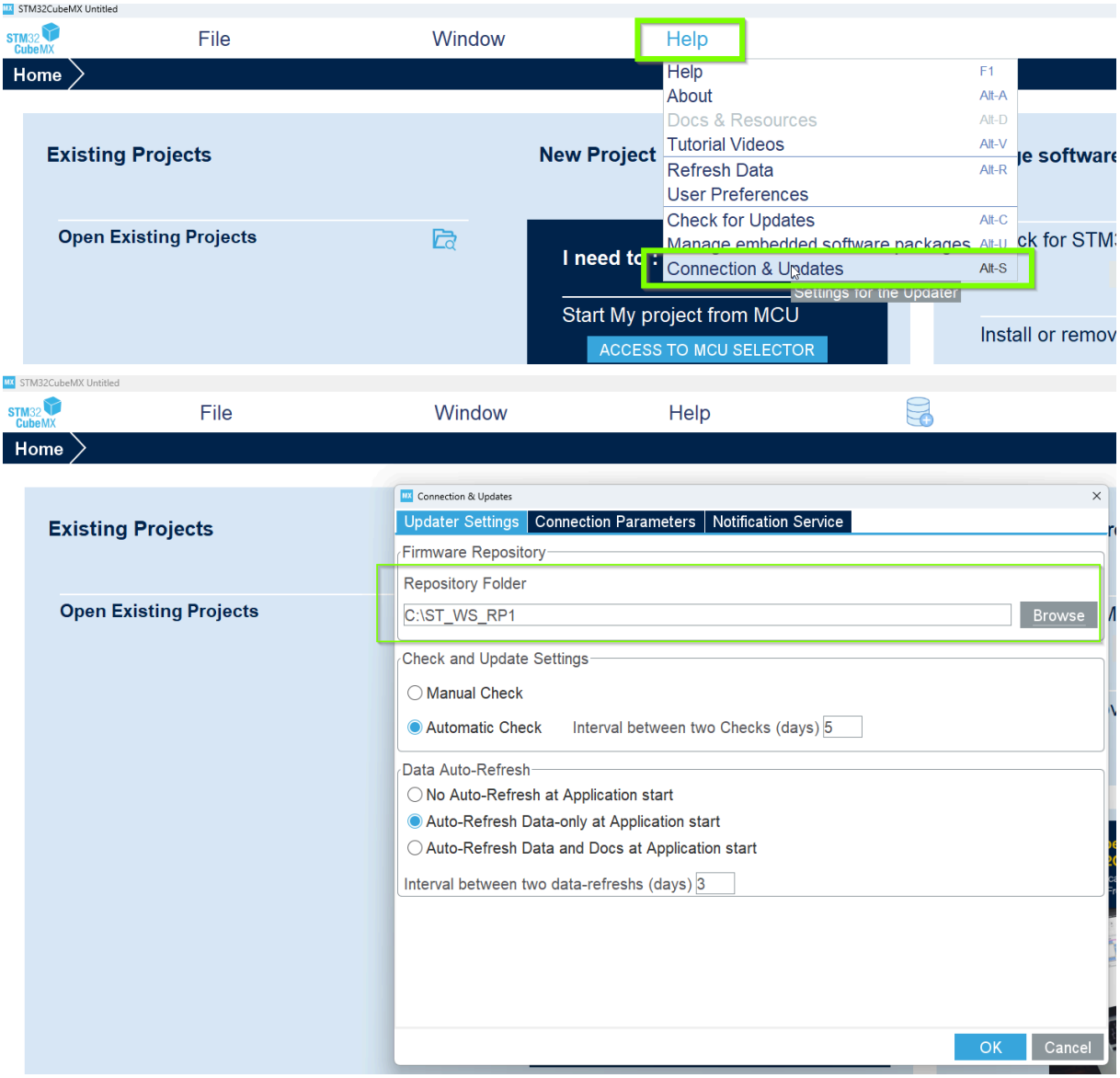
- Crea la carpeta "C:ST\STM32CubeMX-6.18.1", descarga e instala STM32CubeMX(V:6.18.1

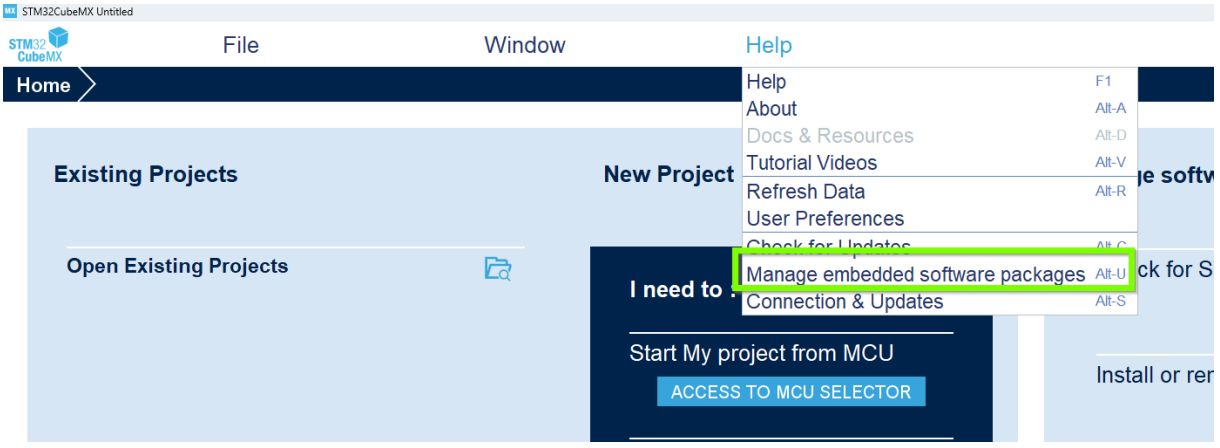


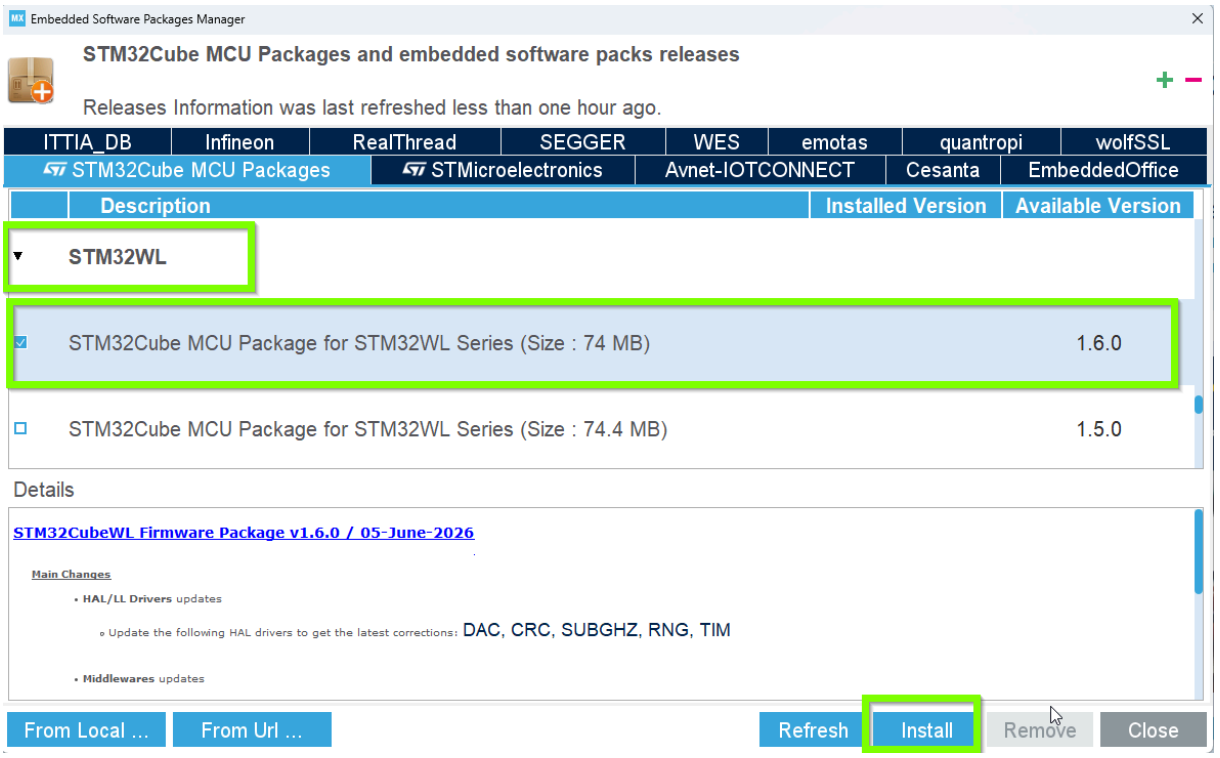

STM32CubeMX Installation Wizard
STM32CubeMX Installation path
MX
life.augmented
STM32 Cube
Select the installation path:
C:\ST\STM32CubeMX-6.18.1
Browse...
STMicroelectronics
Previous
Next
Quit


STM32CubeMX Installation Wizard
STM32CubeMX Shortcuts setup
MX
life.augmented
STM32 Cube
Create shortcuts in the Start-Menu
Create additional shortcuts on the desktop
Select a Program Group for the Shortcuts:
(Default)
Accessibility
Accessories
Administrative Tools
Altium
Brother
Brother P-touch
Business Explorer
CadStd
create shortcut...
current user
all users
STMicroelectronics\STM32Cube\STM32CubeMX
Default
STMicroelectronics
Previous
Next
Quit

- Inicia STM32CubeMX



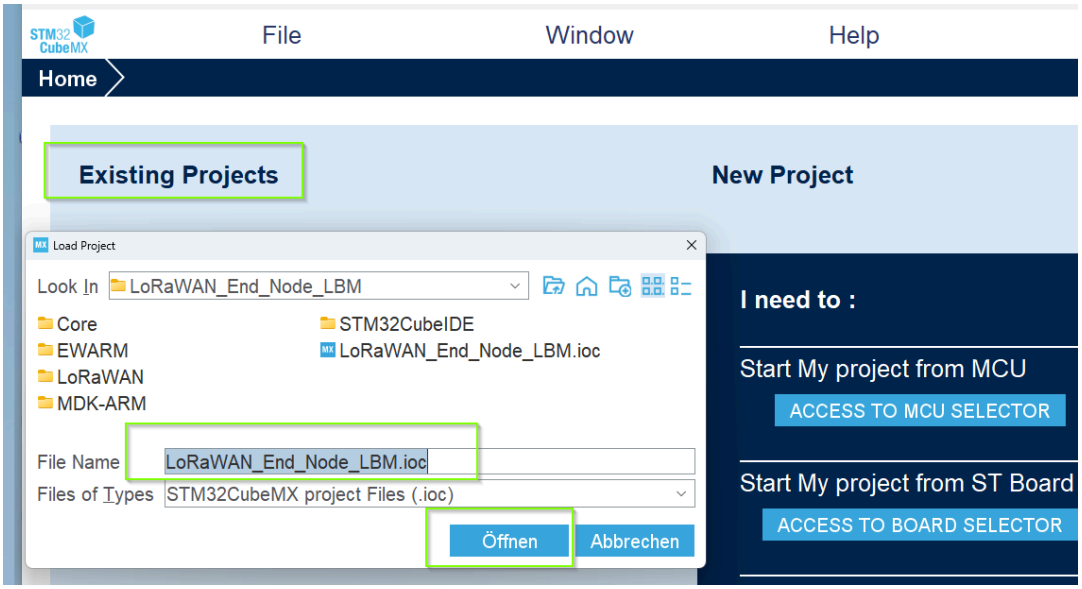

STM32CubeMX Untitled
File
Window
Help
Home
Help F1
About Alt-A
Docs & Resources Alt-D
Tutorial Videos Alt-V
Refresh Data Alt-R
User Preferences
Manage embedded software packages Alt-U
Connection & Updates Alt-S
Existing Projects
New Project
Open Existing Projects
I need to :
Start My project from MCU
ACCESS TO MCU SELECTOR
Install or rer

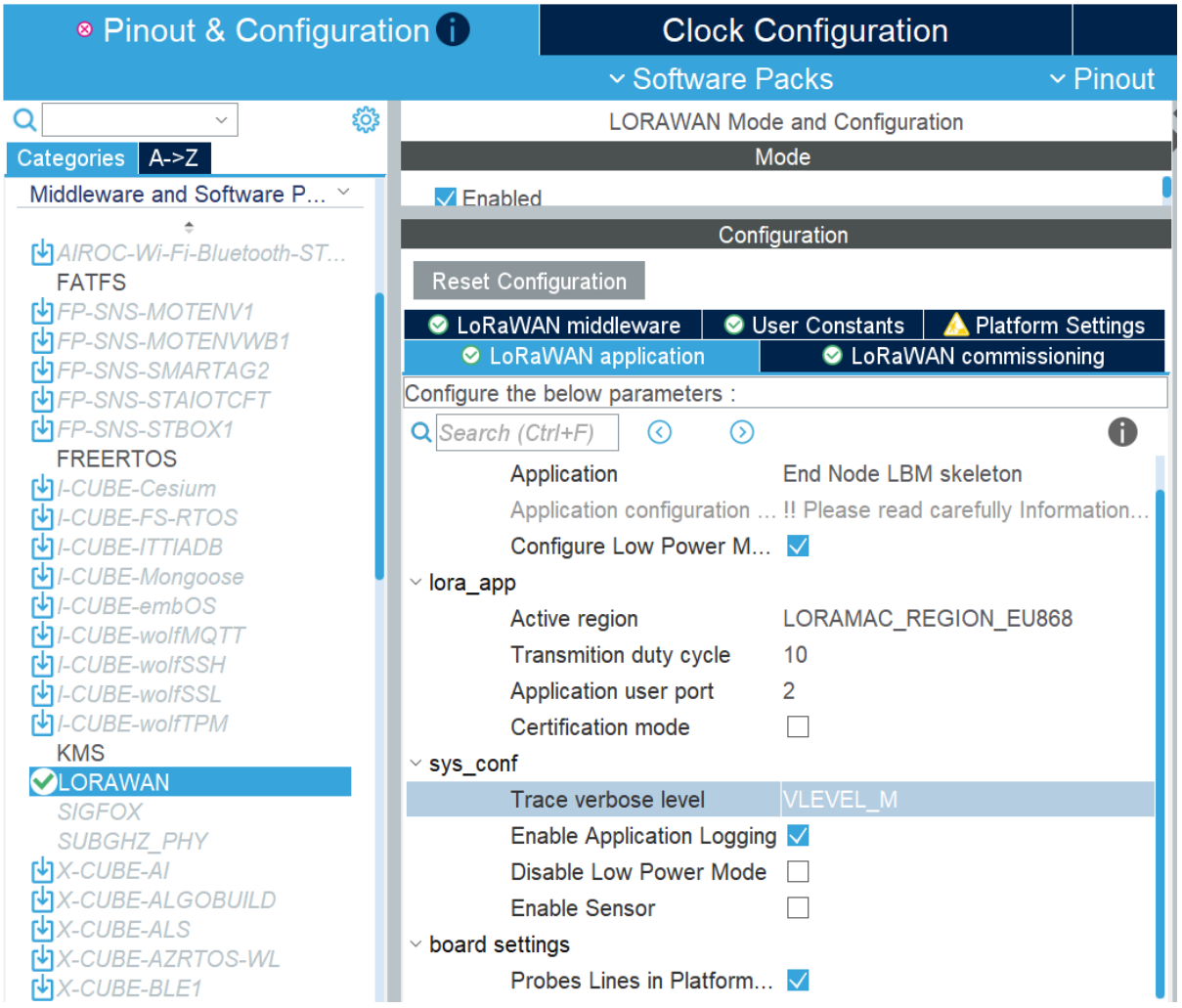

Embedded Software Packages Manager
STM32Cube MCU Packages and embedded software packs releases
Releases Information was last refreshed less than one hour ago.
ITTIA_DB
Infineon
RealThread
SEGGER
WES
emotas
quantropi
wolfSSL
STM32Cube MCU Packages
STMicroelectronics
Avnet-IOTCONNECT
Cesanta
EmbeddedOffice
Description
Installed Version
Available Version
STM32WL
STM32Cube MCU Package for STM32WL Series (Size : 74 MB) 1.6.0
STM32Cube MCU Package for STM32WL Series (Size : 74.4 MB) 1.5.0
Details
STM32CubeWL Firmware Package v1.6.0 / 05-June-2026
Main Changes
HAL/LL Drivers updates
Update the following HAL drivers to get the latest corrections: DAC, CRC, SUBGHZ, RNG, TIM
Middlewares updates
From Local ...
From Url ...
Refresh
Install
Remove
Close

Configuración de LoRaWAN:

- Inicia: LoRaWAN_End_Node_LBM.ioc

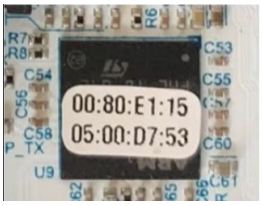
  C:\ST_WS_RP1\STM32Cube_FW_WL_V1.6.0\Proyectos\NUCLEO-WL55JC\applications\LoRaWAN\LoRaWAN_End_Node_LBM\LoRaWAN_End_Node_LBM.ioc





Puesta en marcha de LoRaWAN:

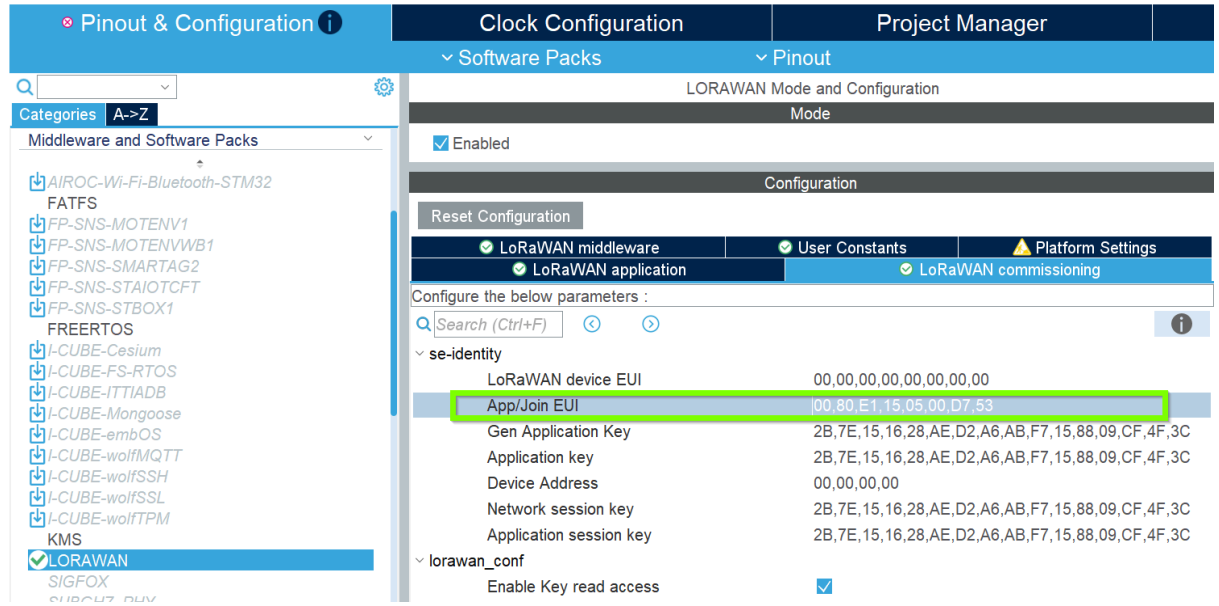
- *“App/Join EUI”*: Se puede encontrar sobre el microcontrolador.

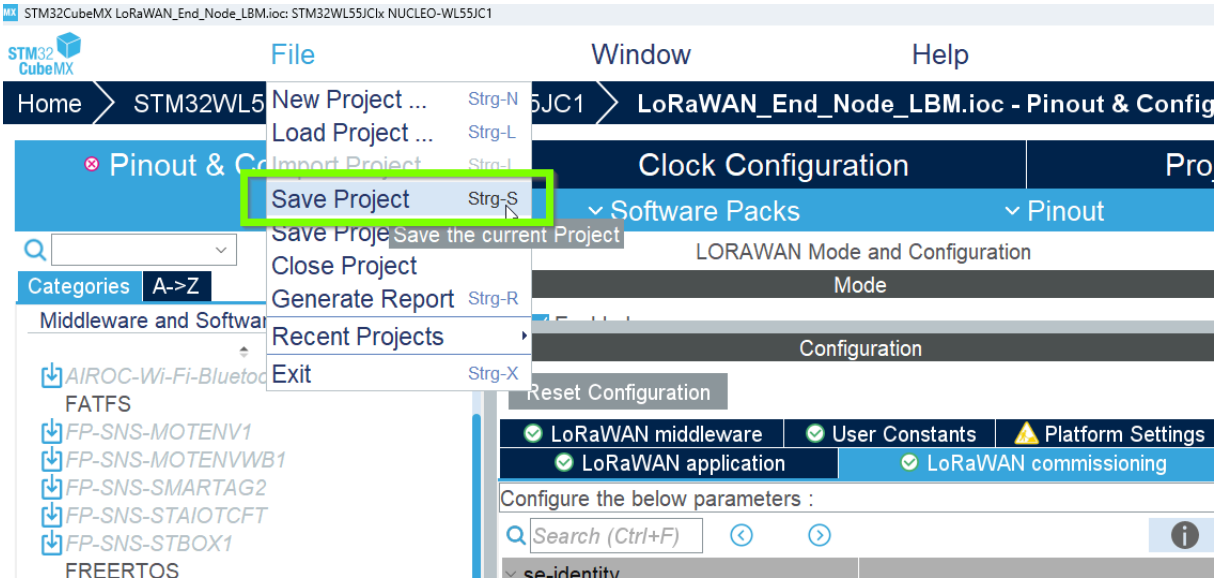


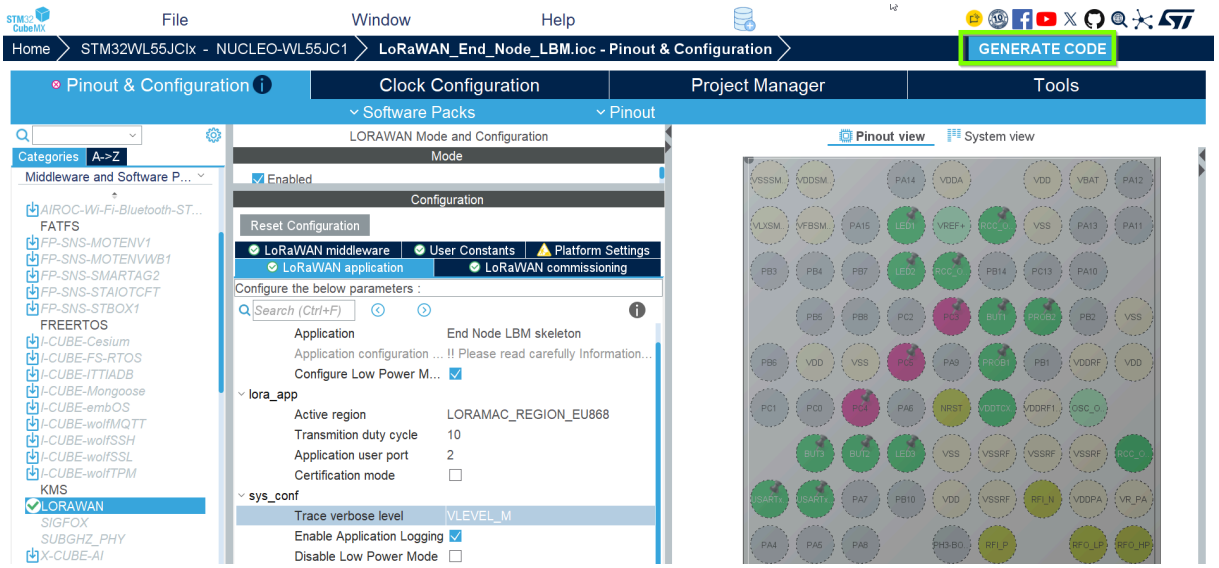
Importante: El campo "LORAWAN_device EUI" NO debe definirse como "00.00.00.00.00.00.00". Elígelo tú mismo.

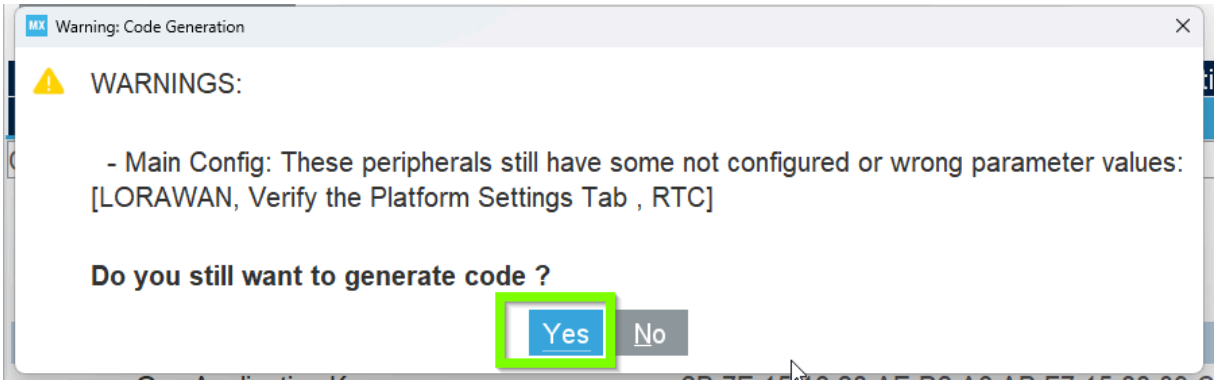
```
/*!
* End-device IEEE EUI (big endian)
* When set to 00,00,00,00,00,00,00,00 DevEui is automatically set
with a value provided by MCU platform
*/
#define LORAWAN_DEVICE_EUI
00,00,00,00,00,00,00,00
```

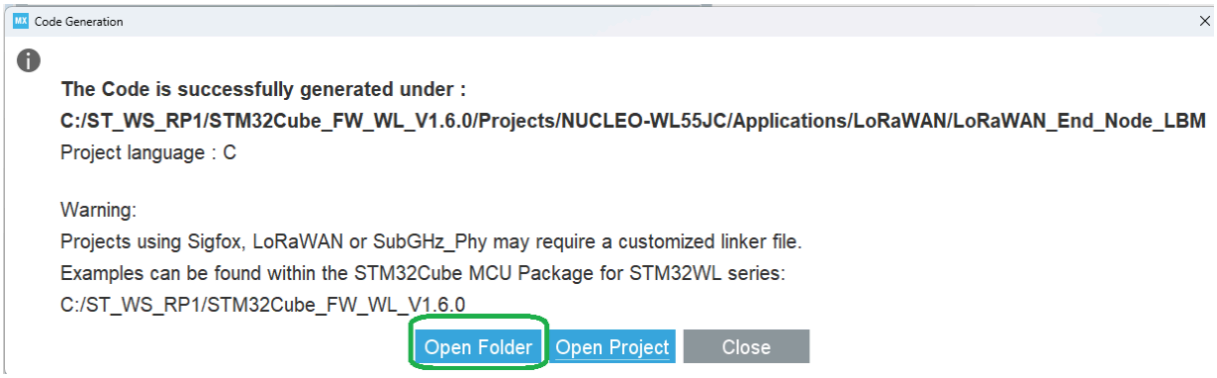
- *“Application key”*: elígelo tú mismo.


Pinout & Configuration
Clock Configuration
Project Manager
Software Packs
Pinout
Categories
A->Z
Middleware and Software Packs
AIROC-Wi-Fi-Bluetooth-STM32
FATFS
FP-SNS-MOTENV1
FP-SNS-MOTENVWB1
FP-SNS-SMARTAG2
FP-SNS-STAIOTCFT
FP-SNS-STBOX1
FREERTOS
I-CUBE-Cesium
I-CUBE-FS-RTOS
I-CUBE-ITTIADB
I-CUBE-Mongoose
I-CUBE-embOS
I-CUBE-wolfMQTT
I-CUBE-wolfSSH
I-CUBE-wolfSSL
I-CUBE-wolfTPM
KMS
LORAWAN
SIGFOX
LORAWAN Mode and Configuration
Mode
Enabled
Configuration
Reset Configuration
LoRaWAN middleware
User Constants
Platform Settings
LoRaWAN application
LoRaWAN commissioning
Configure the below parameters :
Search (Ctrl+F)
se-identity
LoRaWAN device EUI
00,00,00,00,00,00,00,00
App/Join EUI
00,80,E1,15,05,00,D7,53
Gen Application Key
2B,7E,15,16,28,AE,D2,A6,AB,F7,15,88,09,CF,4F,3C
Application key
2B,7E,15,16,28,AE,D2,A6,AB,F7,15,88,09,CF,4F,3C
Device Address
00,00,00,00
Network session key
2B,7E,15,16,28,AE,D2,A6,AB,F7,15,88,09,CF,4F,3C
Application session key
2B,7E,15,16,28,AE,D2,A6,AB,F7,15,88,09,CF,4F,3C
lorawan_conf
Enable Key read access

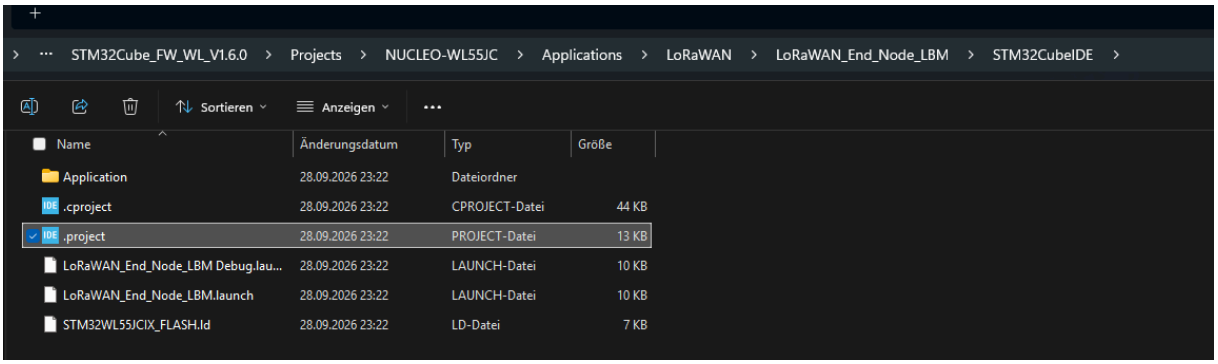

STM32WL55JCIx [NUCLEO-WL55JC1]
File
Window
Help
Home
LoRaWAN_End_Node_LBM.ioc - Pinout & Config
New Project ... Strg-N
Load Project ... Strg-L
Save Project Strg-S
Save the current Project
Close Project
Generate Report Strg-R
Recent Projects
Exit Strg-X
Clock Configuration
Software Packs
Pinout
LORAWAN Mode and Configuration
Mode
Configuration
Reset Configuration
LoRaWAN middleware
User Constants
Platform Settings
LoRaWAN application
LoRaWAN commissioning
Configure the below parameters :
Search (Ctrl+F)


File
Window
Help
Home
STM32WL55JCIx - NUCLEO-WL55JC1
LoRaWAN_End_Node_LBM.ioc - Pinout & Configuration
GENERATE CODE
Pinout & Configuration
Clock Configuration
Project Manager
Tools
Software Packs
Pinout
LORAWAN Mode and Configuration
Mode
Enabled
Configuration
Reset Configuration
LoRaWAN middleware
User Constants
Platform Settings
LoRaWAN application
LoRaWAN commissioning
Configure the below parameters :
Search (Ctrl+F)
Application
End Node LBM skeleton
Application configuration ...
!! Please read carefully Information...
Configure Low Power M...
lora_app
Active region
LORAMAC_REGION_EU868
Transmition duty cycle
10
Application user port
2
Certification mode
sys_conf
Trace verbose level
VLEVEL_M
Enable Application Logging
Disable Low Power Mode
Pinout view
System view

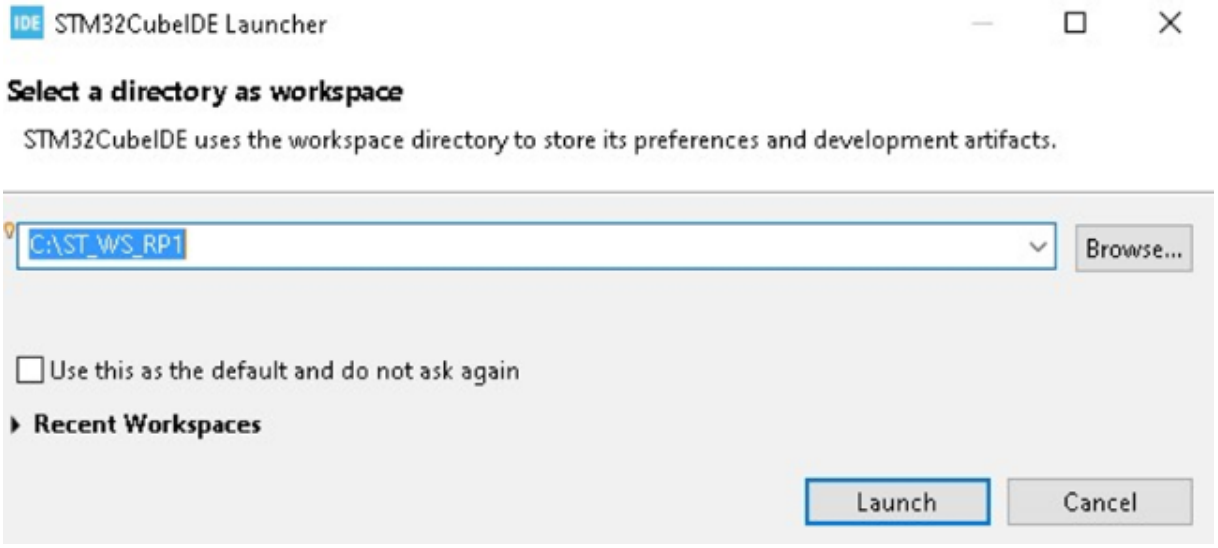


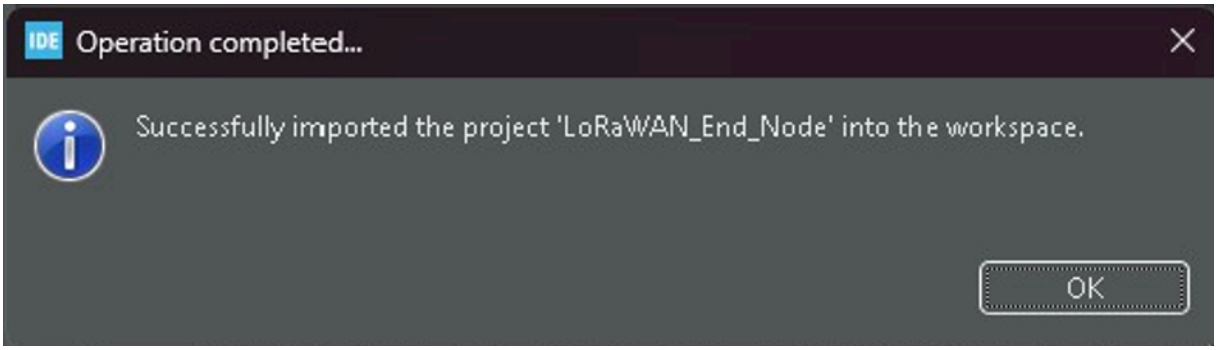


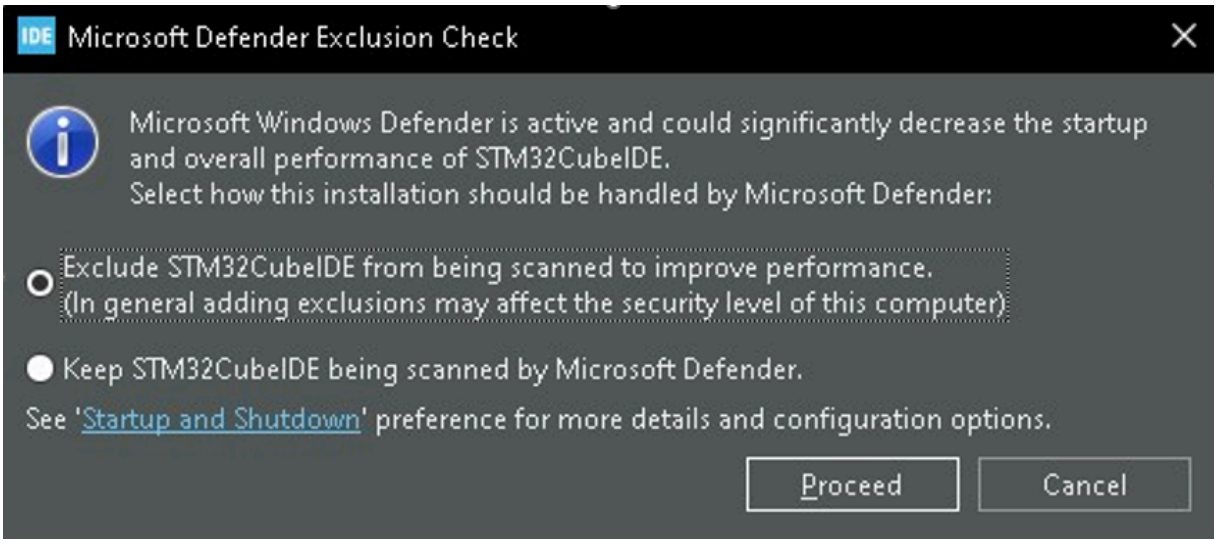
- Inicie el proyecto con STM32cubeIDE.
"*C:\ST_WS_RP1\STM32Cube_FW_WL_V1.6.0\Projects\NUCLEO-WL55JC\Applications\LoRaWAN\LoRaWAN_End_Node_LBM\STM32CubeIDE\.project*"

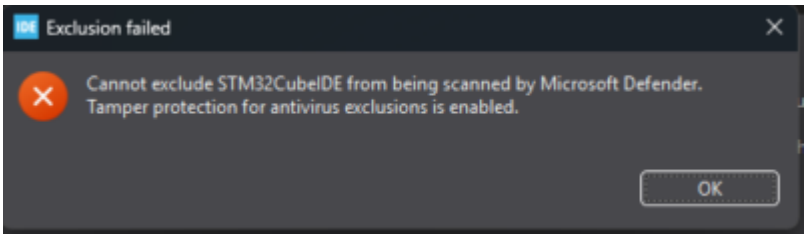


- Seleccione C:\ST_WS_RP1 como espacio de trabajo.

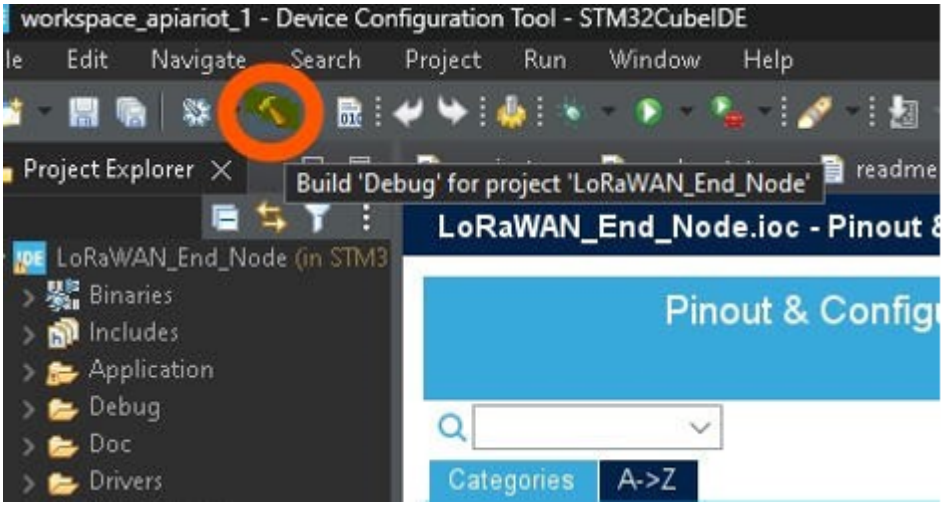


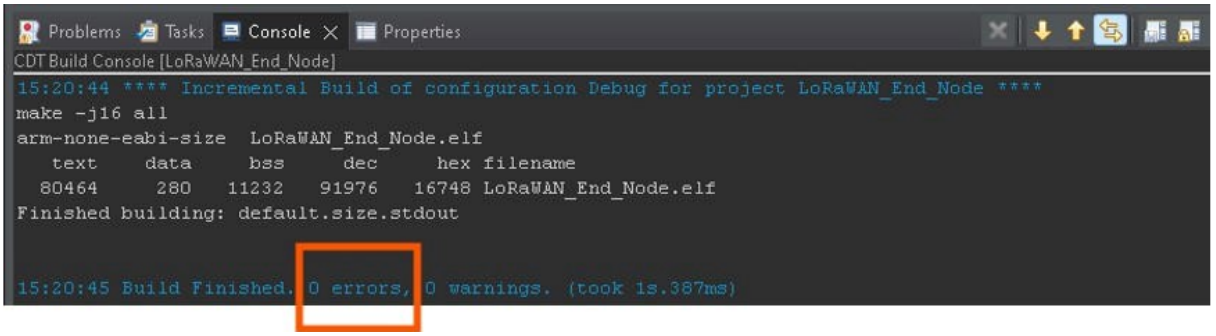

Operation completed...
Successfully imported the project 'LoRaWAN_End_Node' into the workspace.
OK


Microsoft Defender Exclusion Check
Microsoft Windows Defender is active and could significantly decrease the startup and overall performance of STM32CubeIDE.
Select how this installation should be handled by Microsoft Defender:
Exclude STM32CubeIDE from being scanned to improve performance.
(In general adding exclusions may affect the security level of this computer)
Keep STM32CubeIDE being scanned by Microsoft Defender.
See 'Startup and Shutdown' preference for more details and configuration options.
Proceed
Cancel


Exclusion failed
Cannot exclude STM32CubeIDE from being scanned by Microsoft Defender.
Tamper protection for antivirus exclusions is enabled.
OK

- Compila el proyecto sin errores:





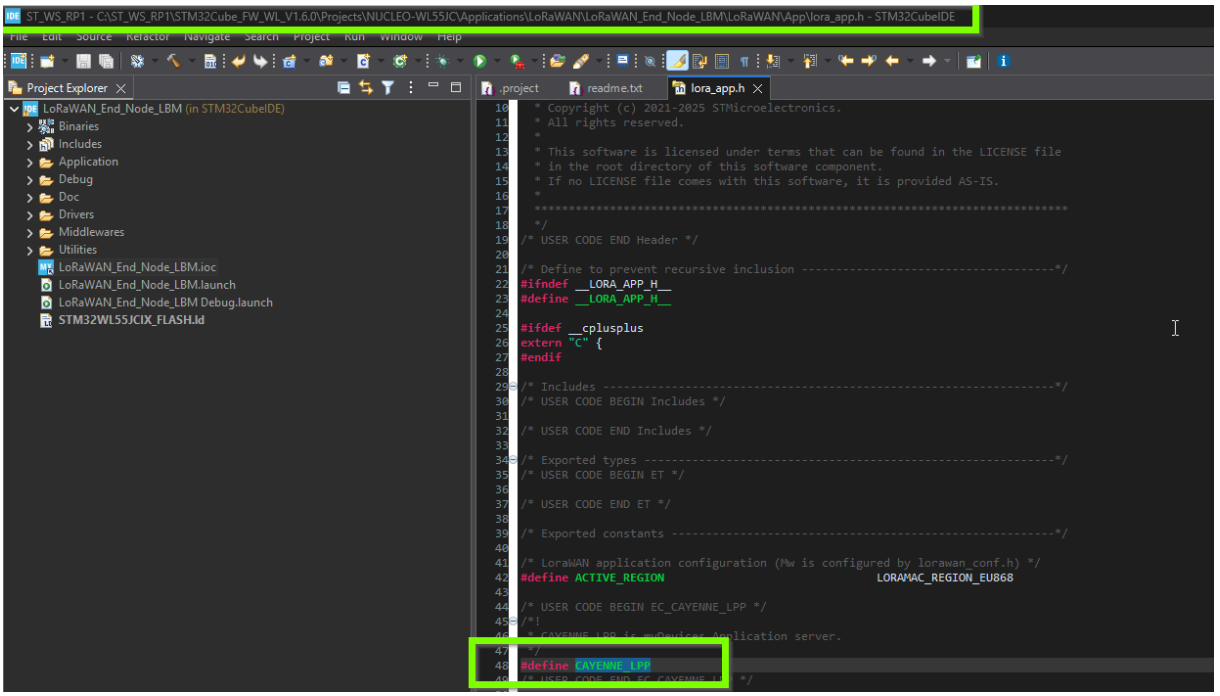
Cambio de código de programación

- Activa la “Low Power Payload” de Cayenne:
    - Descomente /*#define CAYENNE_LPP en “lora_app.h”*/:
      #define CAYENNE_LPP

"C:\ST_WS_RP1\STM32Cube_FW_WL_V1.6.0\Projects\NUCLEO-WL55JC\Applications\LoRaWAN\LoRaWAN_End_Node_LBM\LoRaWAN\App\lora_app.h"

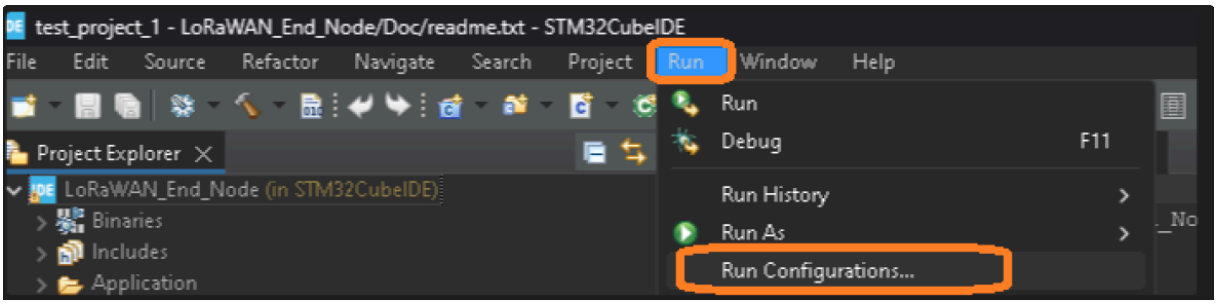
Un aviso: Cayenne LPP es un formato de datos compacto para codificar datos de sensores para LoRaWAN. Puede encontrar más información en este enlace:

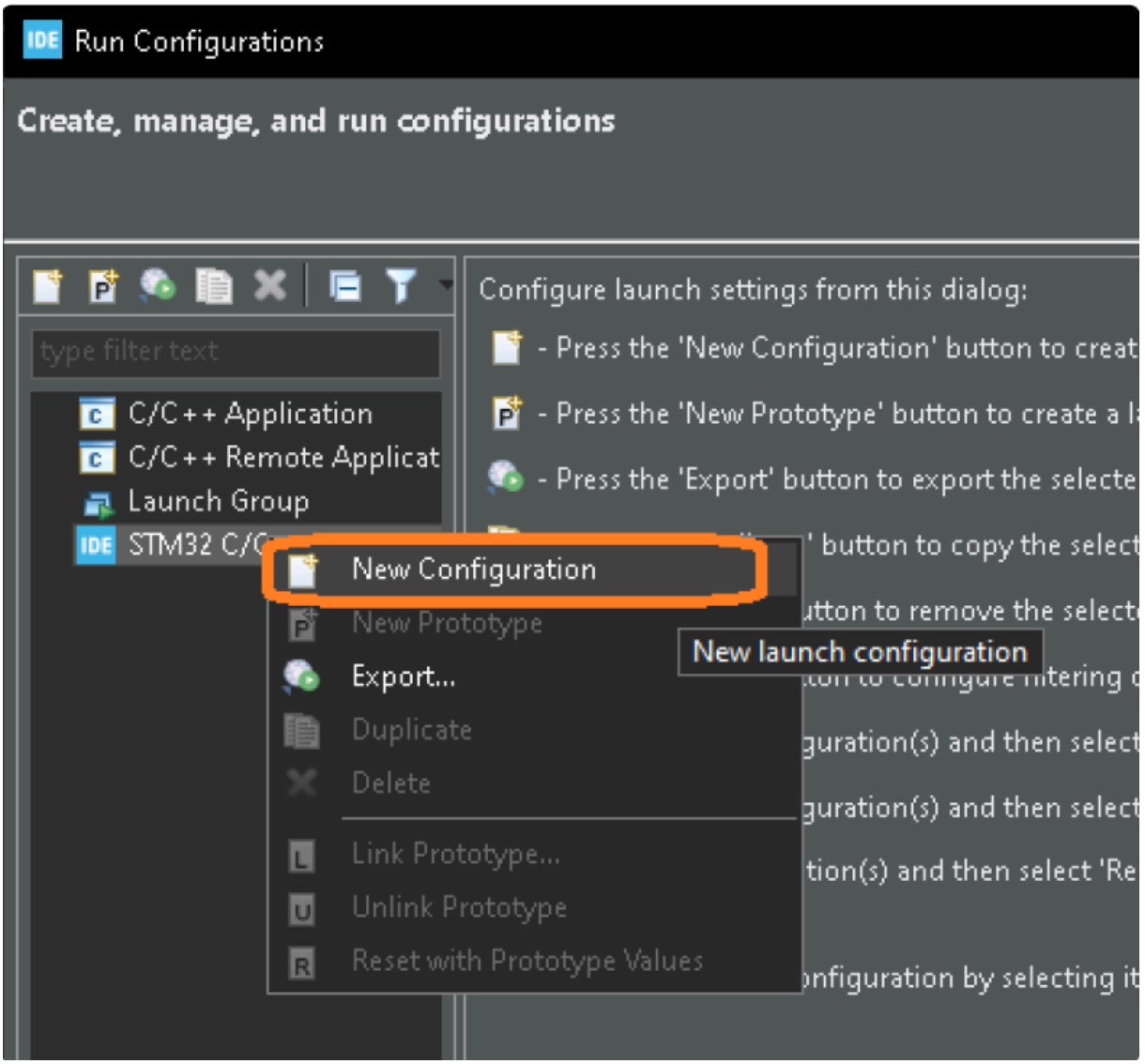
¿Que es “Cayenne Low Power Payload”? 🤔🤷



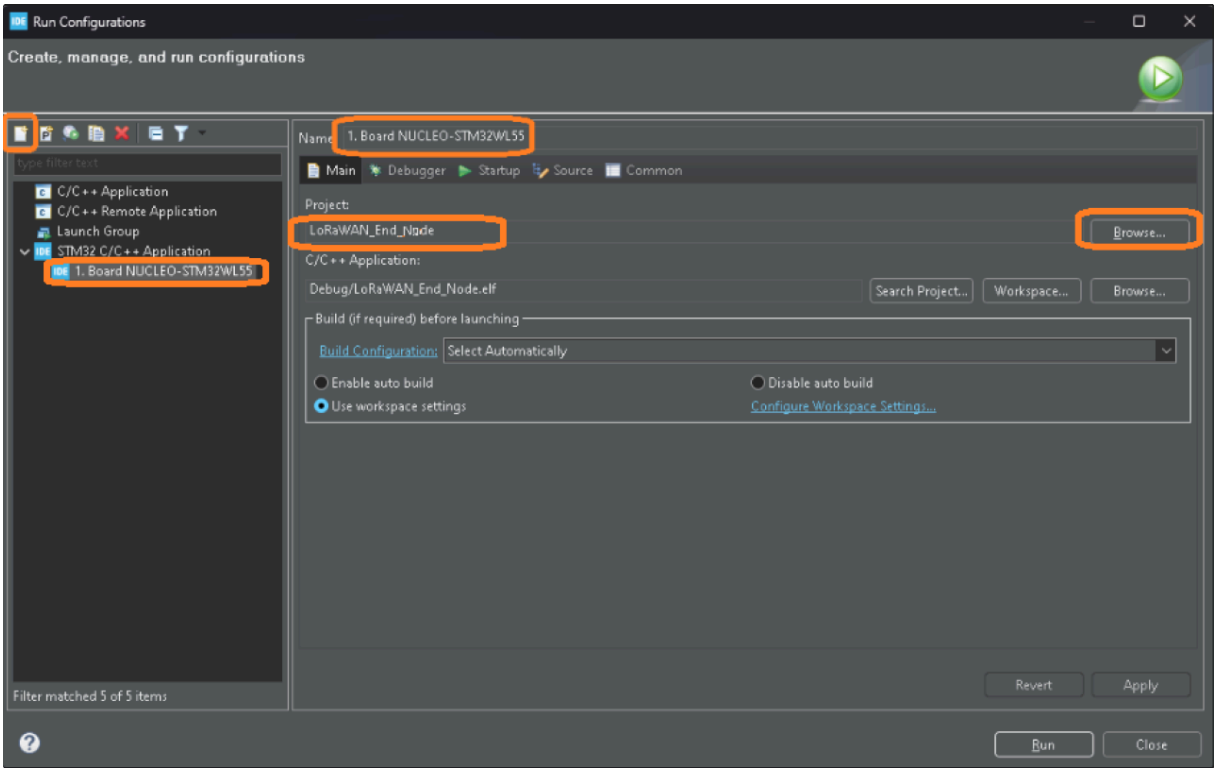
Configuración de la Placa

- Crea una nueva configuración para la placa de desarrollo.

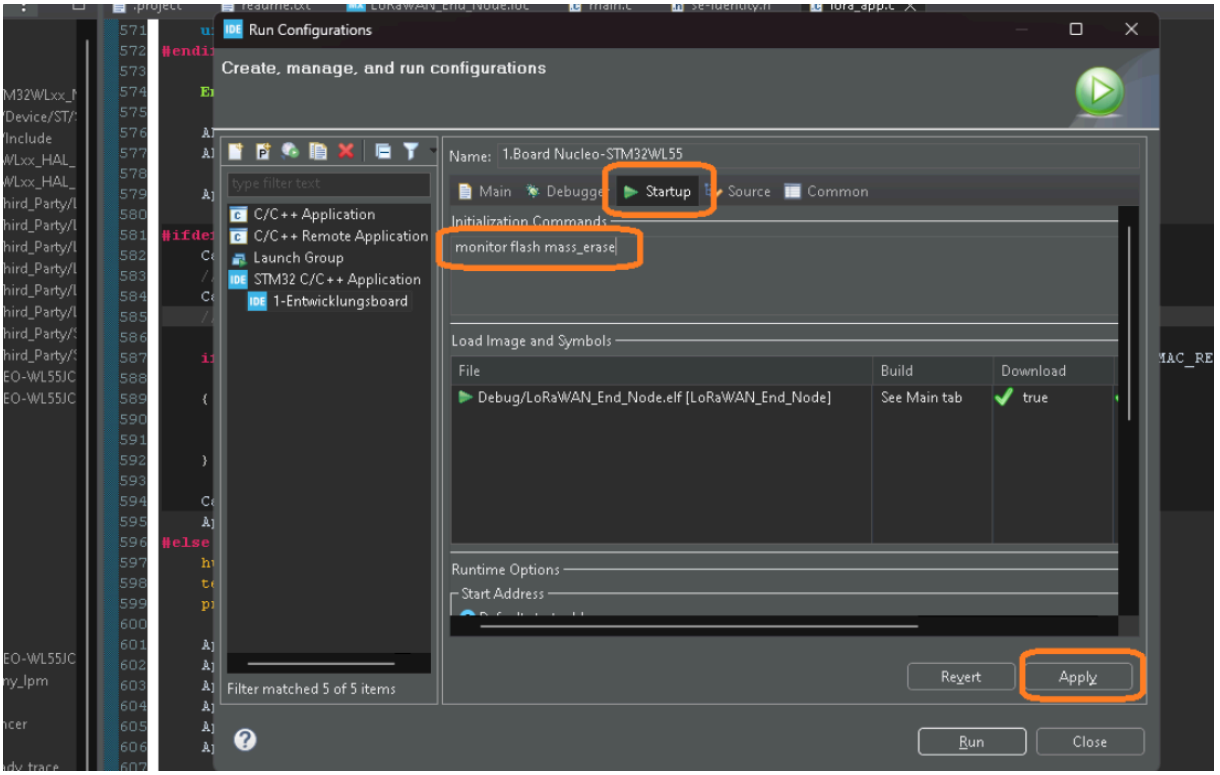




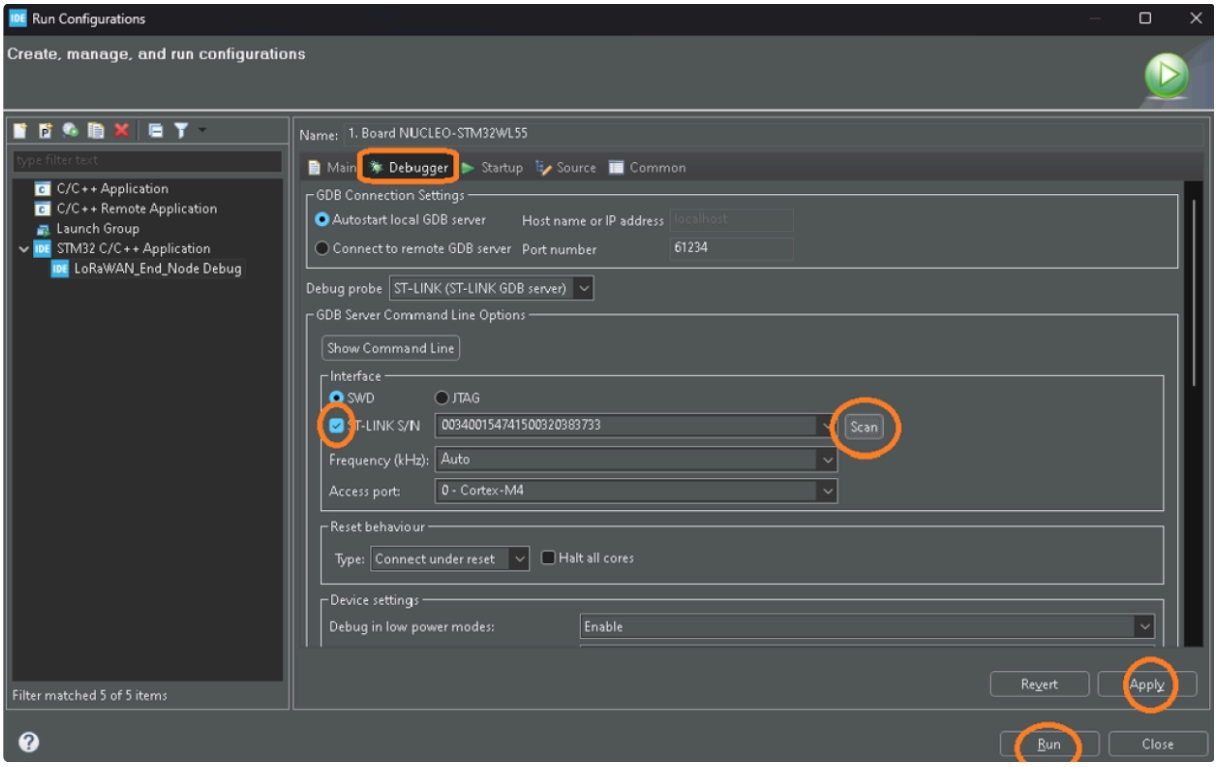
- Nombra tu placa NUCLEO-STM32WL55



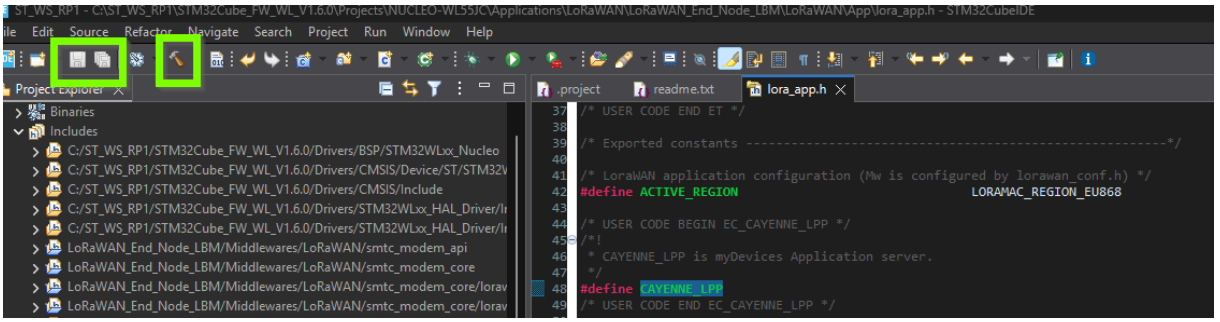
- Agrega en "*Initialization Commands"* el siguiente comando:
  "*monitor flash mass_erase"*



- Conecte la placa de desarrollo al PC mediante USB.

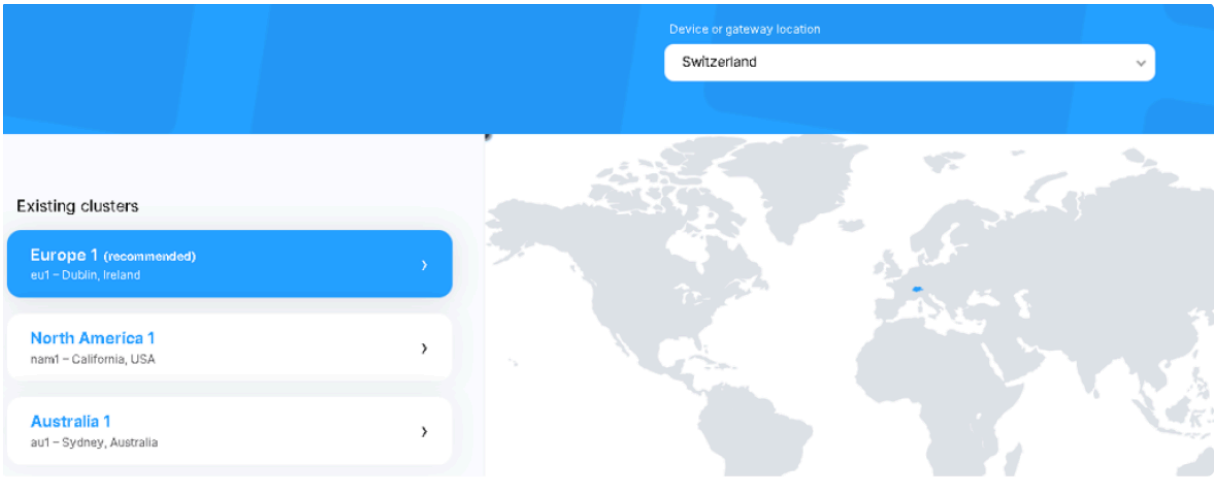


- Guardar la configuración



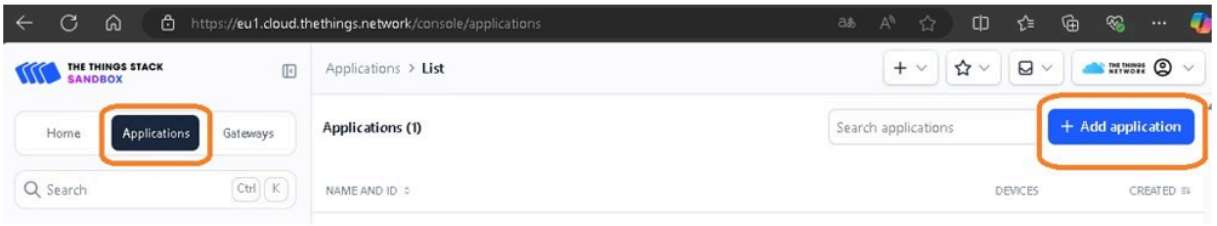
“The Thing Network” (TTN) / “The Stacks”

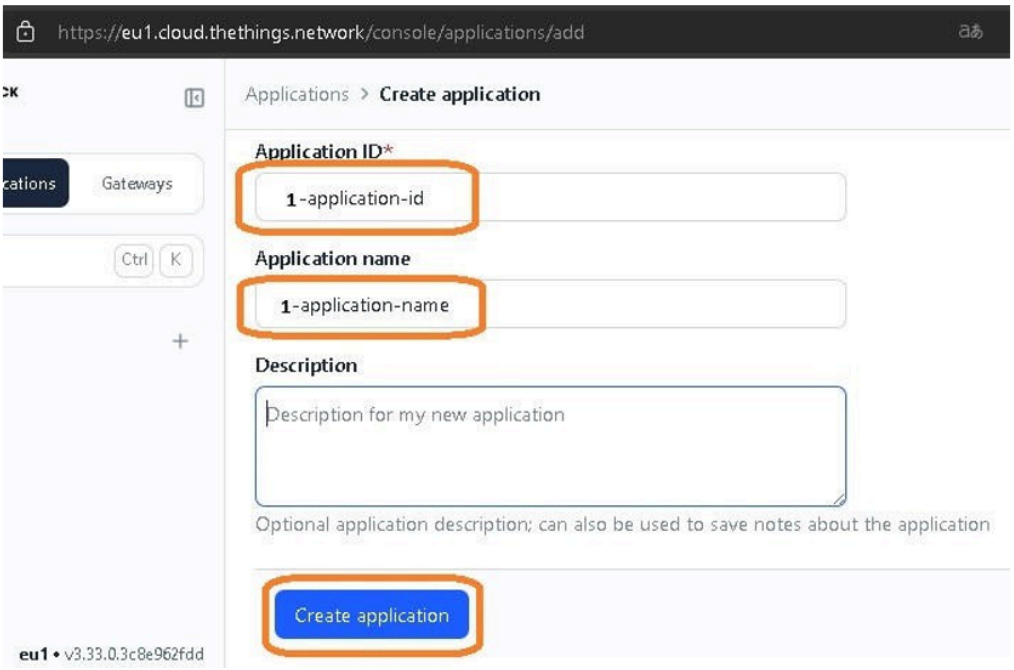
- Crear una cuenta para acceder a ”The things network”.
- Seleccione el clúster Europa 1 (para Suiza)

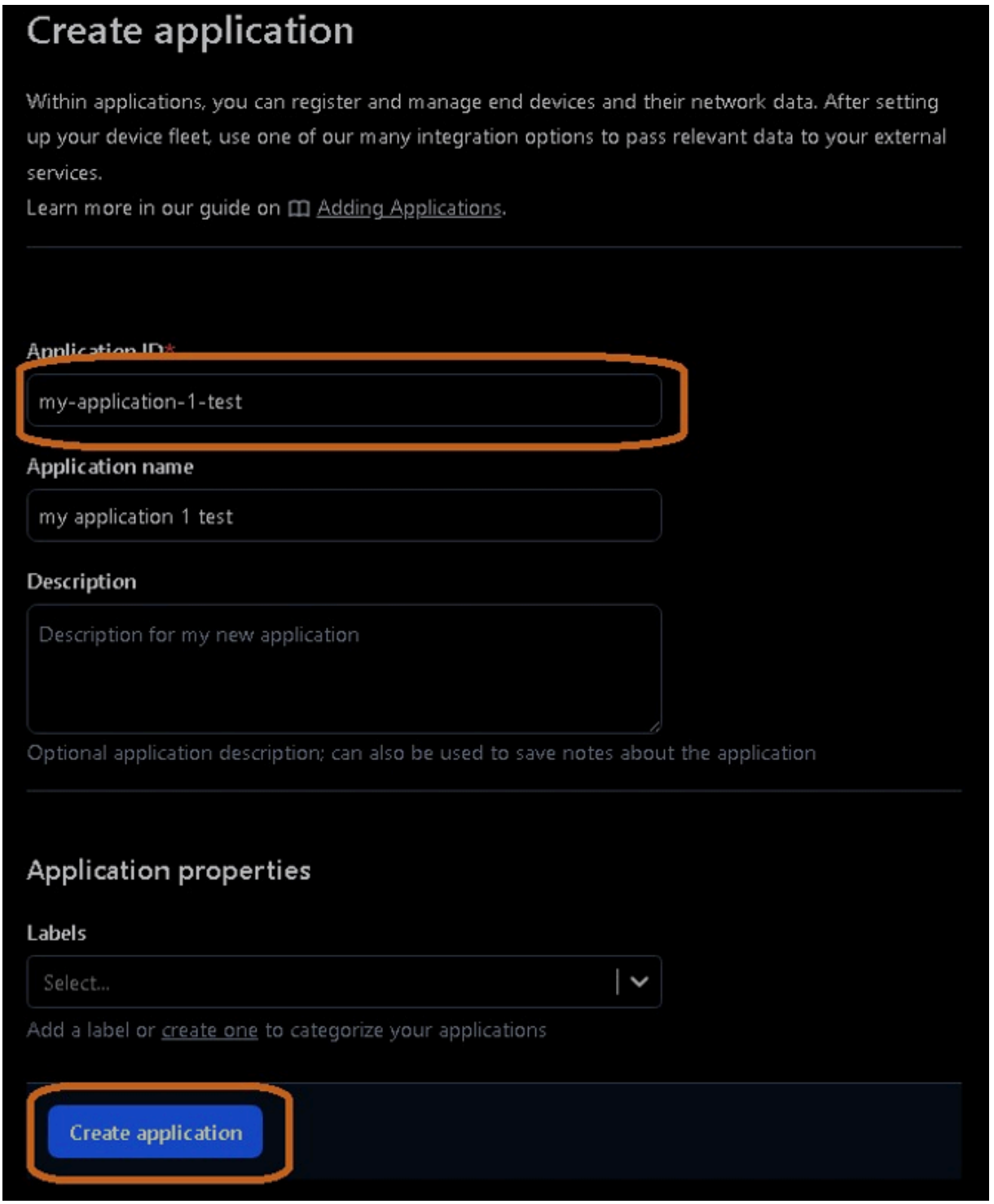


Cluster Für die Schweiz: Europe 1

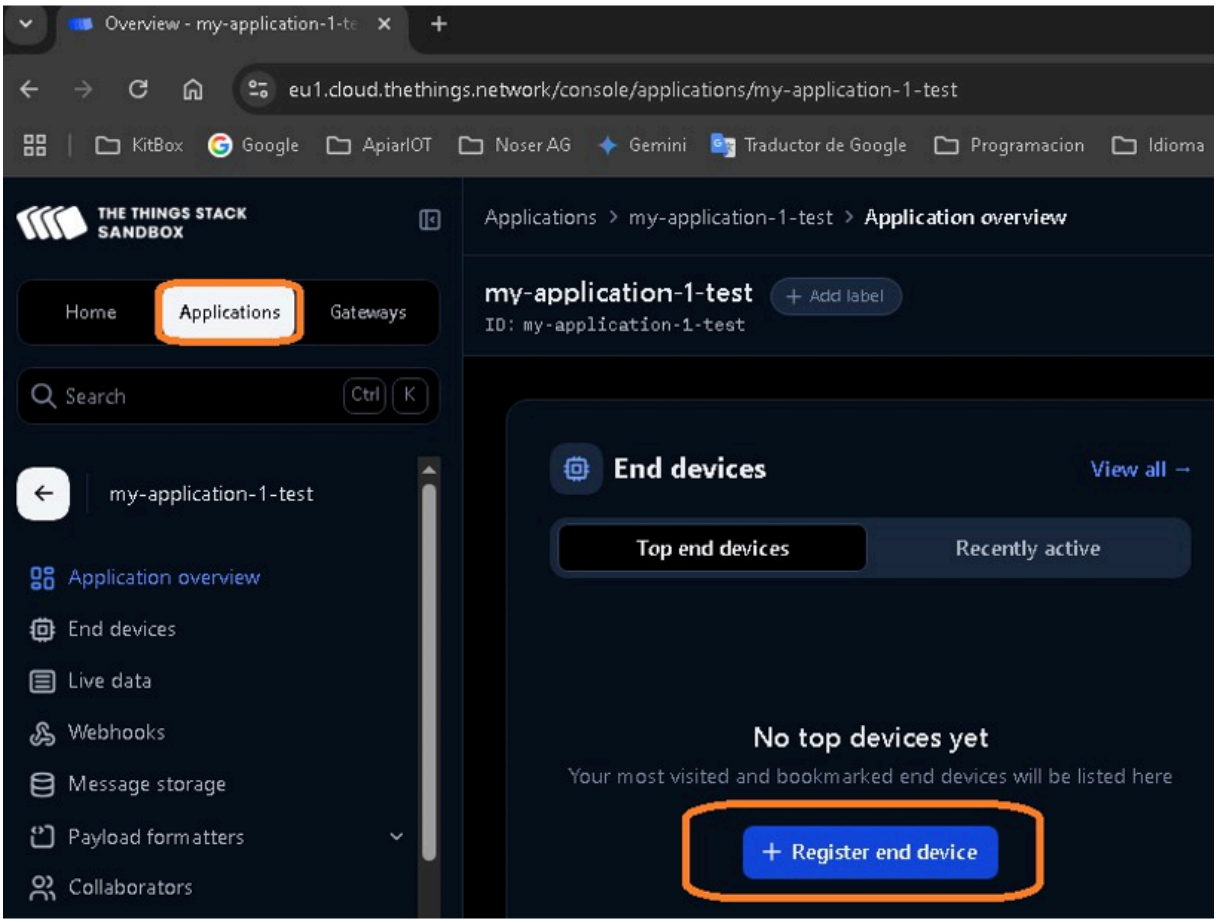
- Crear una aplicación



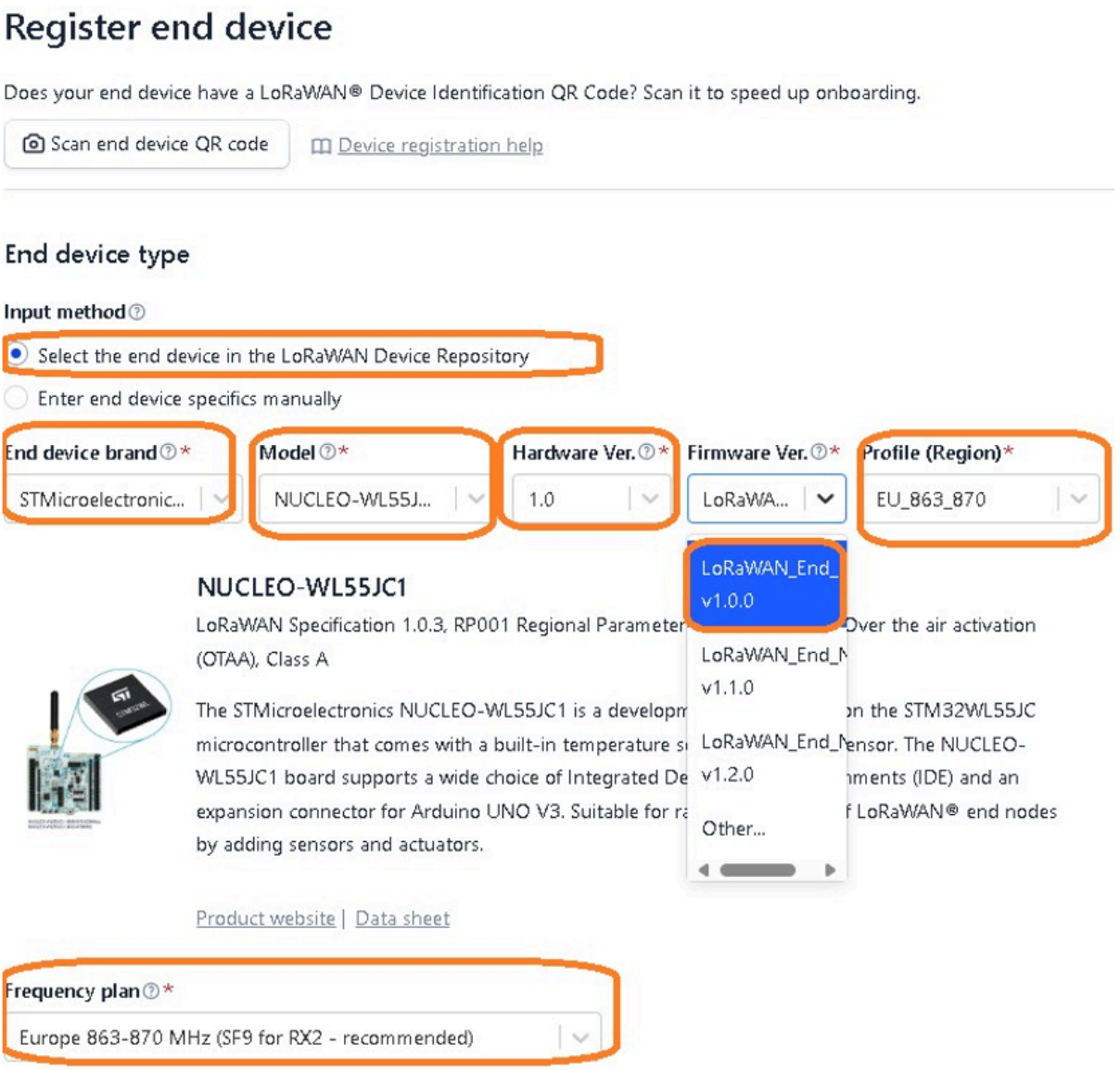



Create application
Within applications, you can register and manage end devices and their network data. After setting up your device fleet, use one of our many integration options to pass relevant data to your external services.
Learn more in our guide on Adding Applications.
Application ID*
my-application-1-test
Application name
my application 1 test
Description
Description for my new application
Optional application description; can also be used to save notes about the application
Application properties
Labels
Select...
Add a label or create one to categorize your applications
Create application

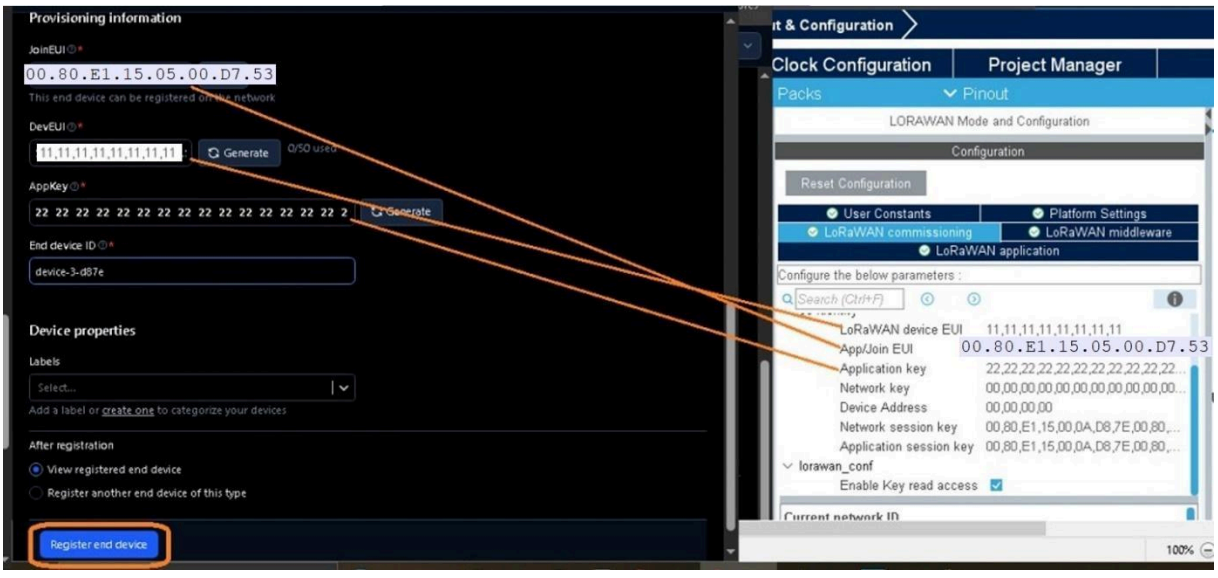
- Registra el "end device".

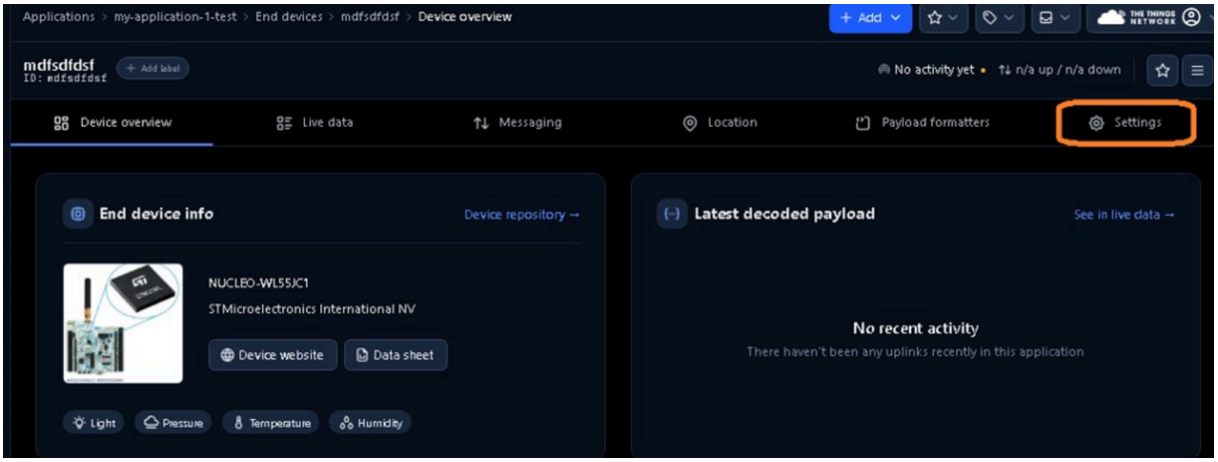


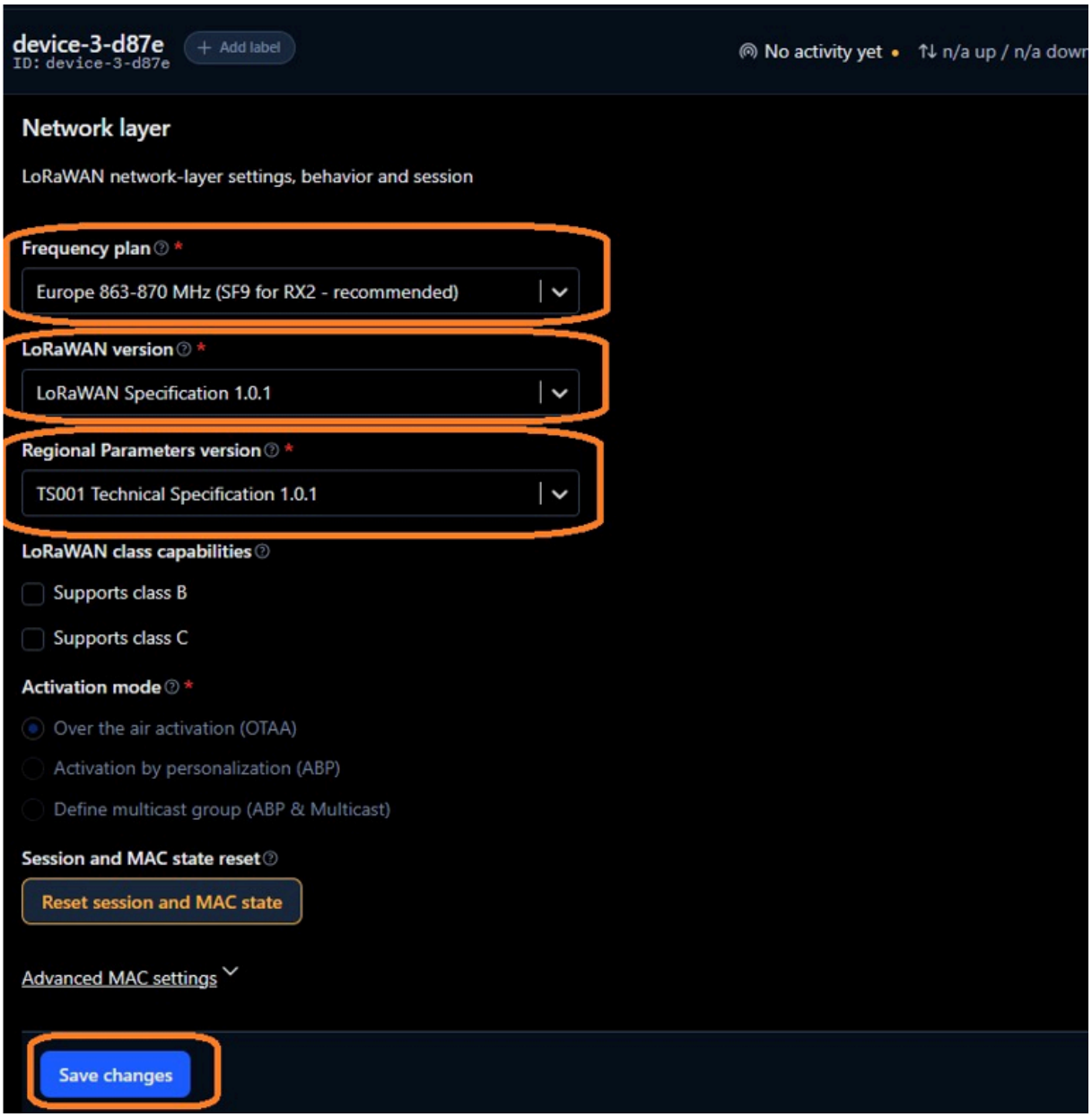
- Tipo de "end device", Marca del "end device", Modelo, Versión de hardware, Versión de firmware, Región del perfil, Plan de frecuencia



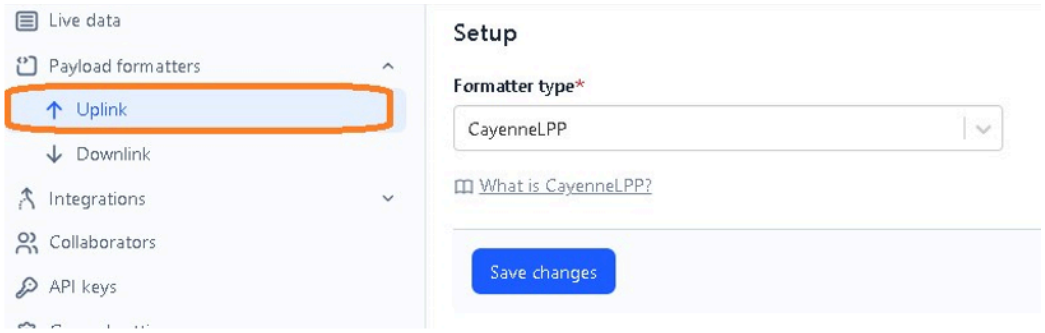
- Buscar la información en "LoRaWAN commissioning"

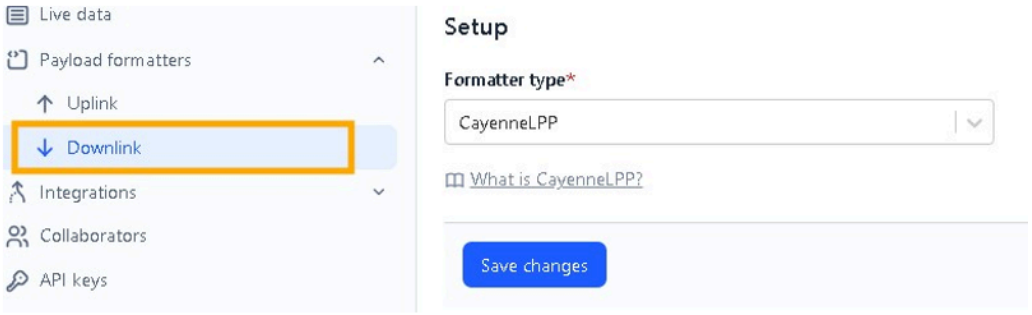



Applications > my-application-1-test > End devices > mdfsdfdsf > Device overview
+ Add
mdfsdfdsf
ID: mdfsdfdsf
+ Add label
No activity yet • ↑↓ n/a up / n/a down
Device overview
Live data
Messaging
Location
Payload formatters
Settings
End device info
Device repository →
NUCLEO-WL55JC1
STMicroelectronics International NV
Device website
Data sheet
Light
Pressure
Temperature
Humidity
Latest decoded payload
See in live data →
No recent activity
There haven't been any uplinks recently in this application

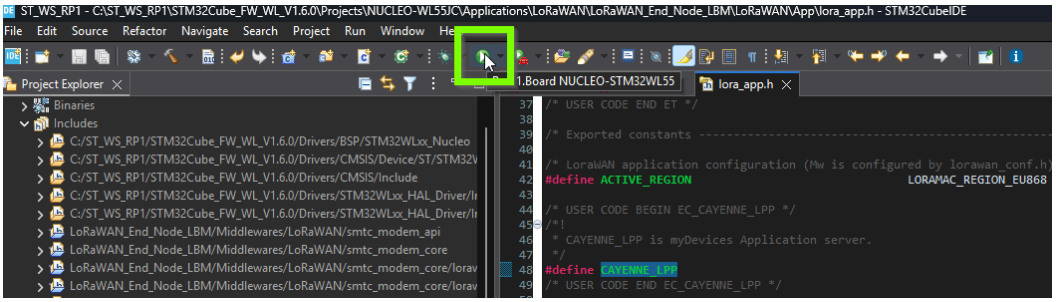

device-3-d87e
ID: device-3-d87e
+ Add label
No activity yet • ↑↓ n/a up / n/a down
Network layer
LoRaWAN network-layer settings, behavior and session
Frequency plan *
Europe 863-870 MHz (SF9 for RX2 - recommended)
LoRaWAN version *
LoRaWAN Specification 1.0.1
Regional Parameters version *
TS001 Technical Specification 1.0.1
LoRaWAN class capabilities
Supports class B
Supports class C
Activation mode *
Over the air activation (OTAA)
Activation by personalization (ABP)
Define multicast group (ABP & Multicast)
Session and MAC state reset
Reset session and MAC state
Advanced MAC settings
Save changes

- Configure el "Uplink" y el "Downlink" para la "Payload del "End-device".

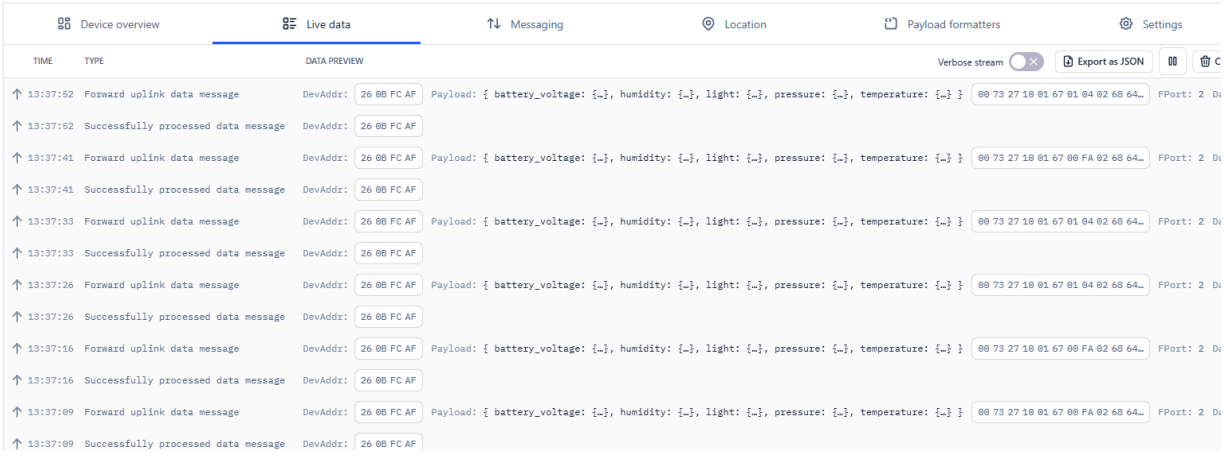




- Sube el programa al tablero.

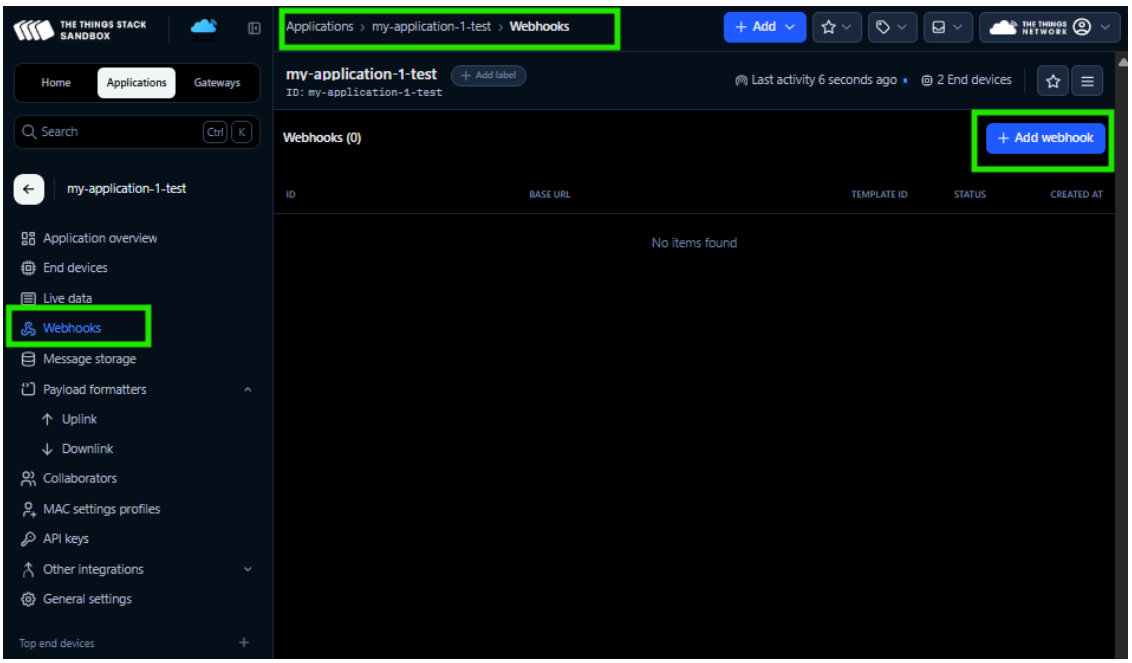


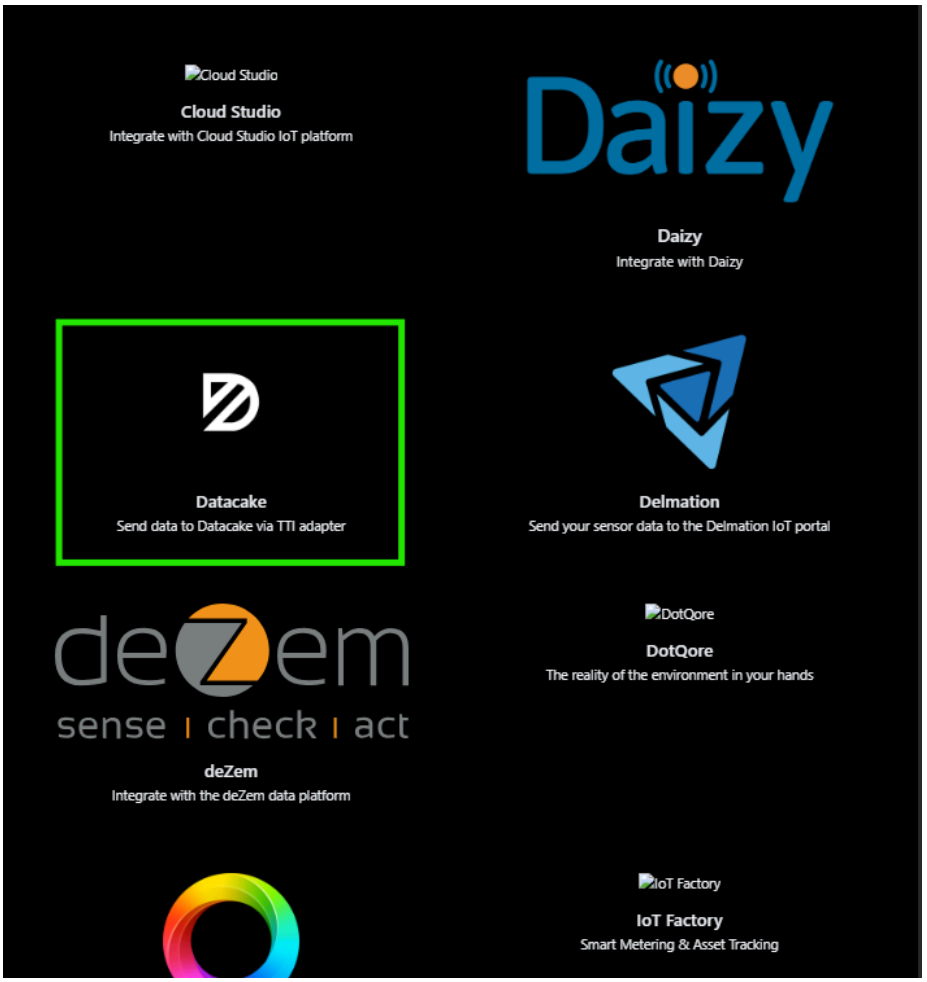
- Compruebe si TTN está recibiendo los datos; de ser así, apague la placa de desarrollo.

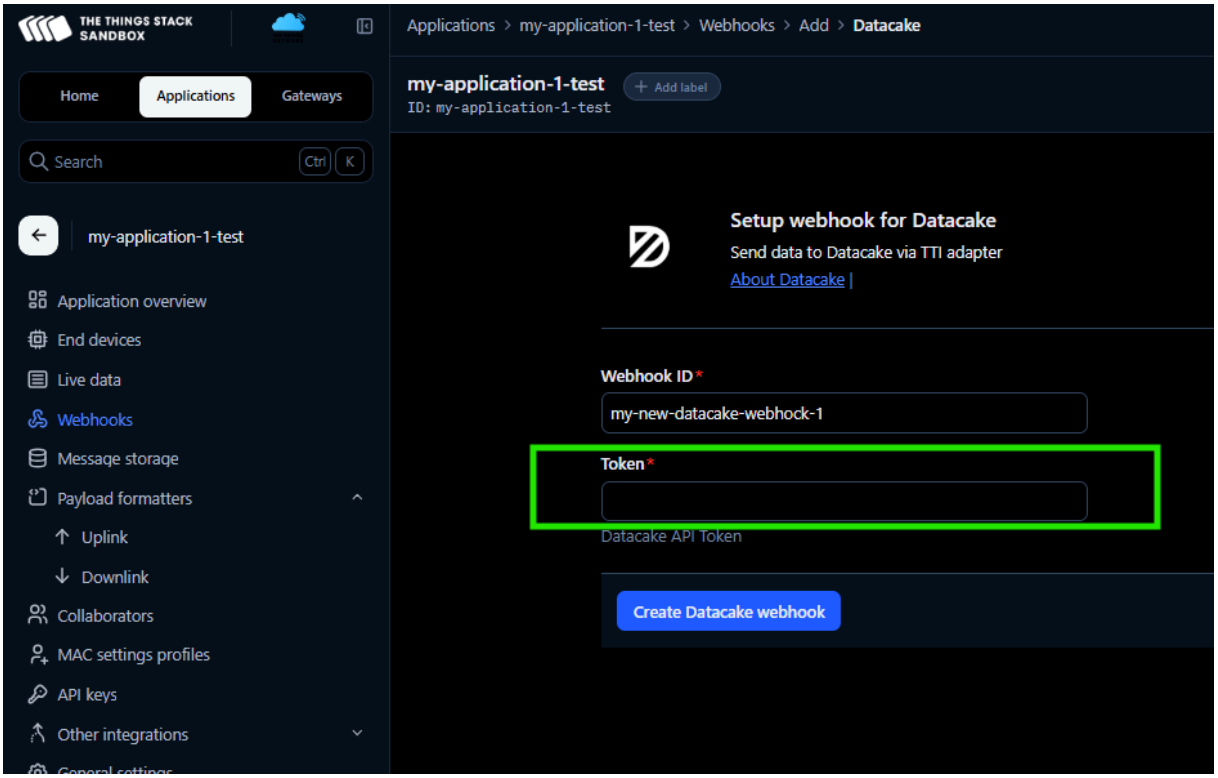


Reenvío de enlace ascendente desde TTN a Datacake

- Configurar webhooks

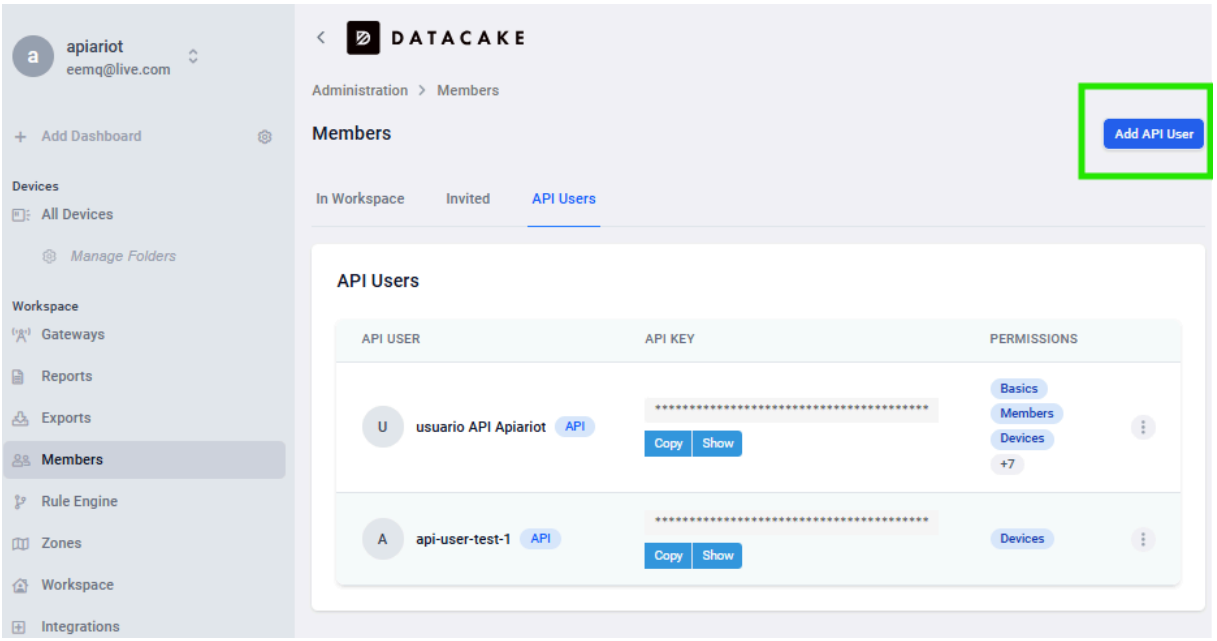


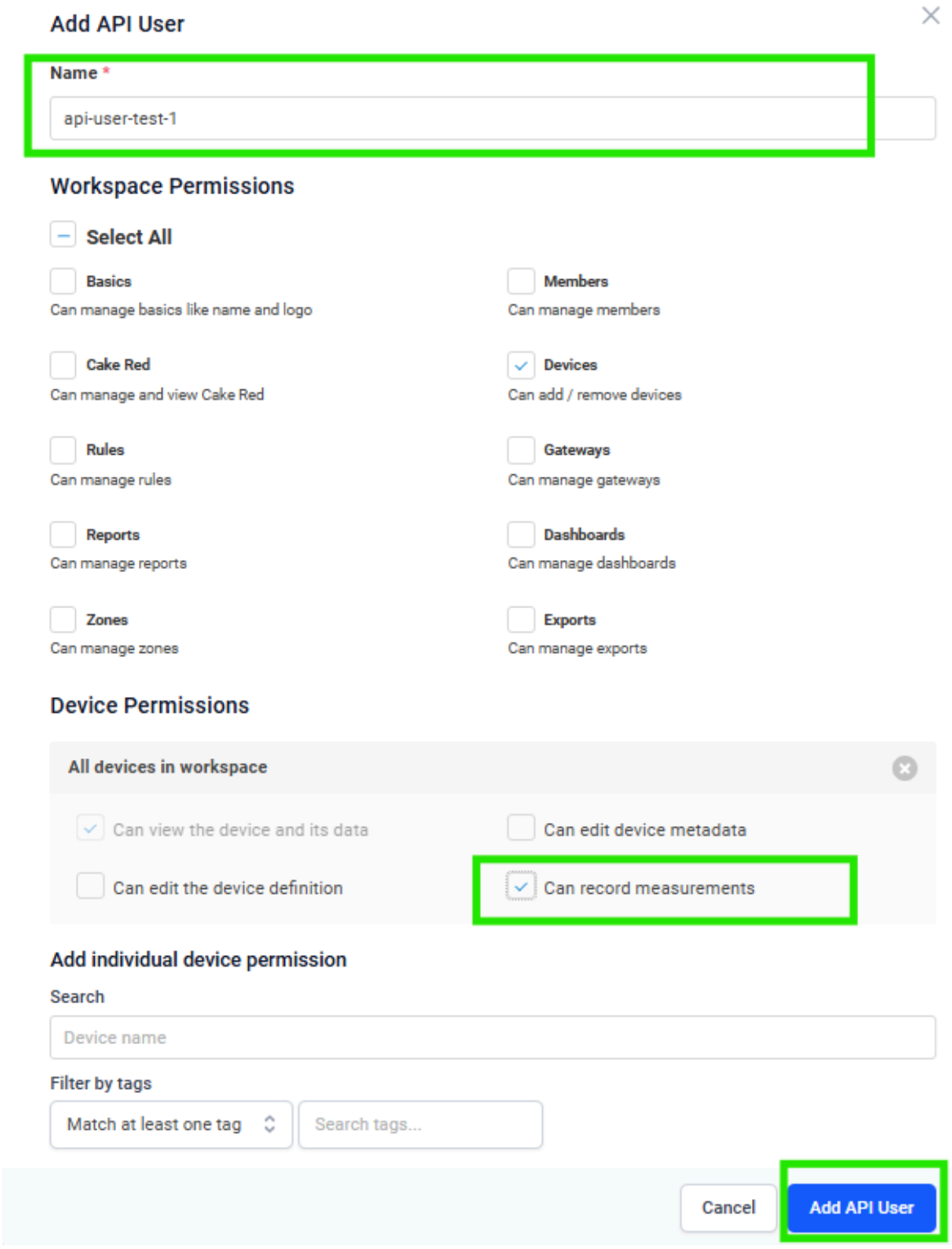




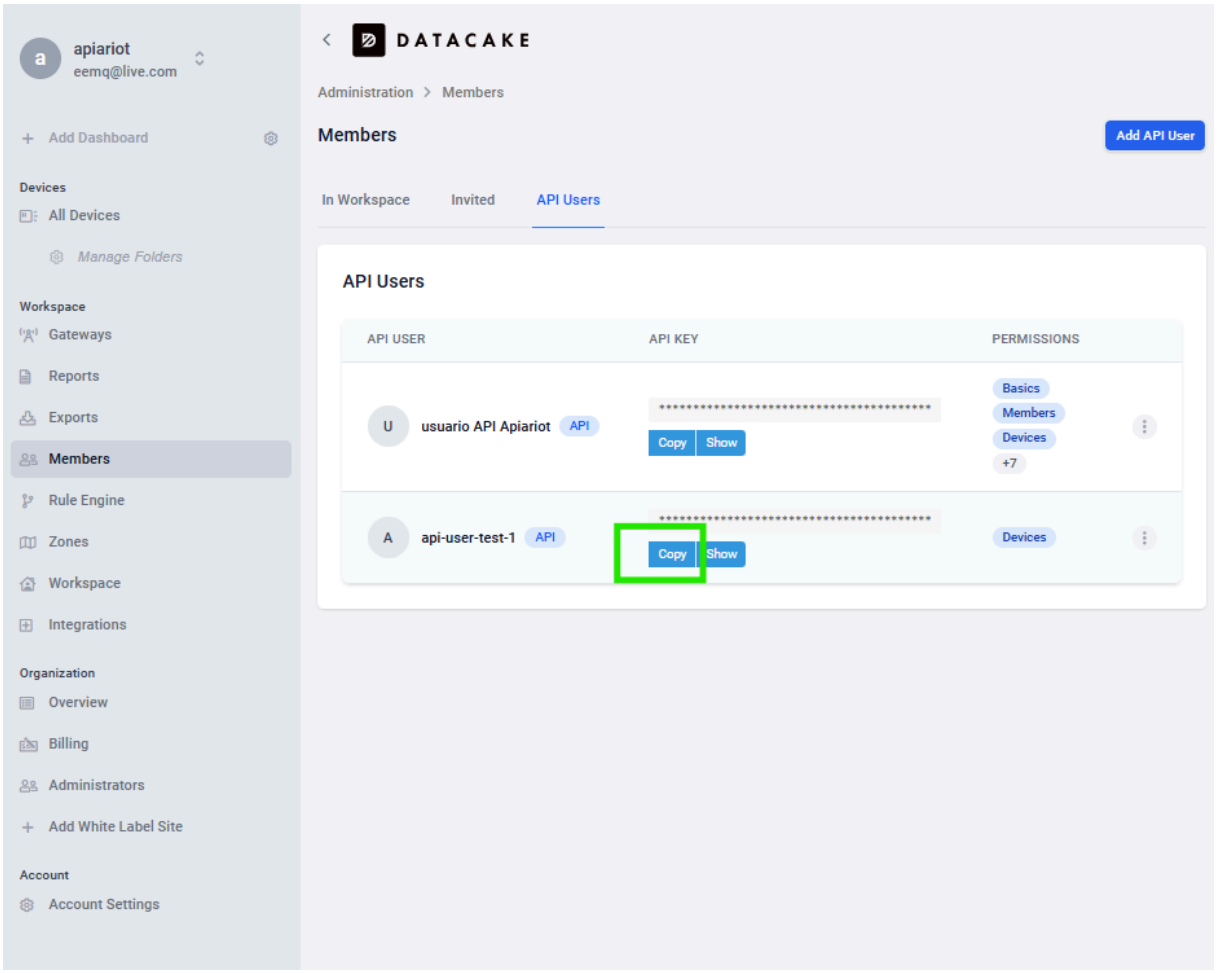
- Primero necesitas una cuenta de Datacake para obtener el token.
- Crea tu cuenta en Datacake.

- Y añade un "Usuario de API".

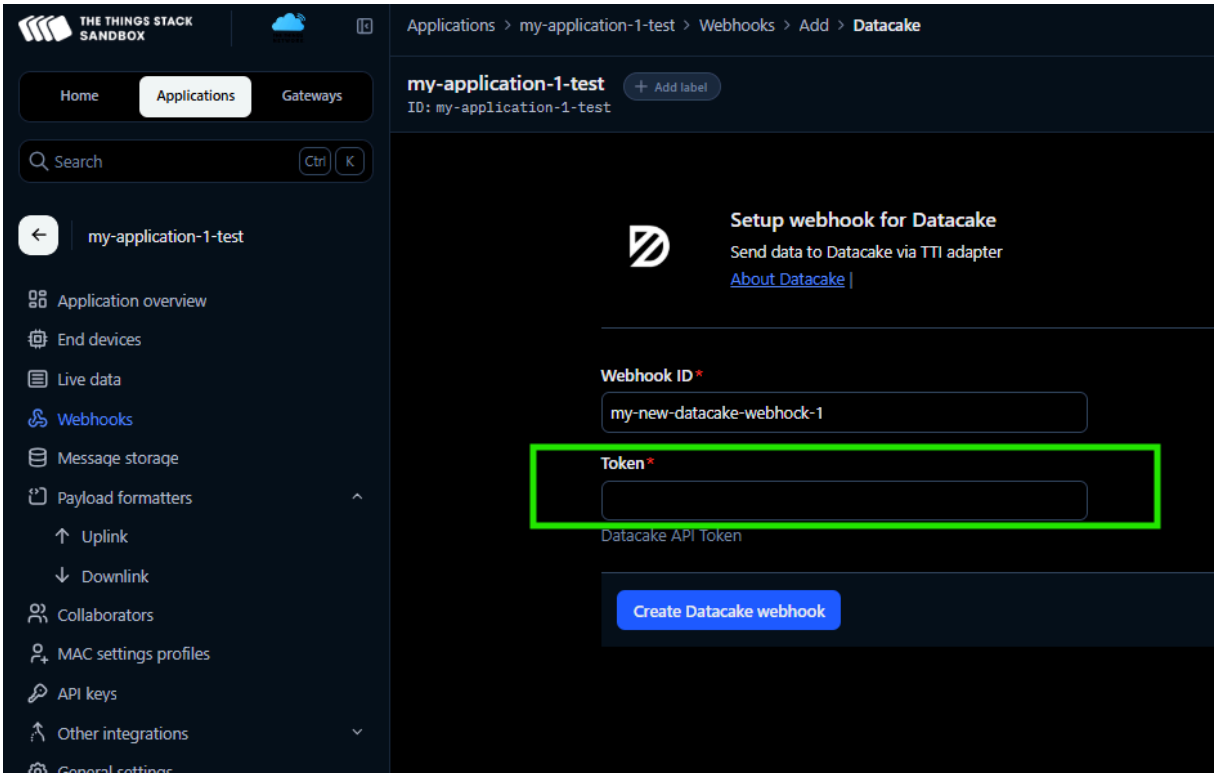



Add API User
Name *
api-user-test-1
Workspace Permissions
Select All
Basics
Can manage basics like name and logo
Members
Can manage members
Cake Red
Can manage and view Cake Red
Devices
Can add / remove devices
Rules
Can manage rules
Gateways
Can manage gateways
Reports
Can manage reports
Dashboards
Can manage dashboards
Zones
Can manage zones
Exports
Can manage exports
Device Permissions
All devices in workspace
Can view the device and its data
Can edit device metadata
Can edit the device definition
Can record measurements
Add individual device permission
Search
Device name
Filter by tags
Match at least one tag
Search tags...
Cancel
Add API User

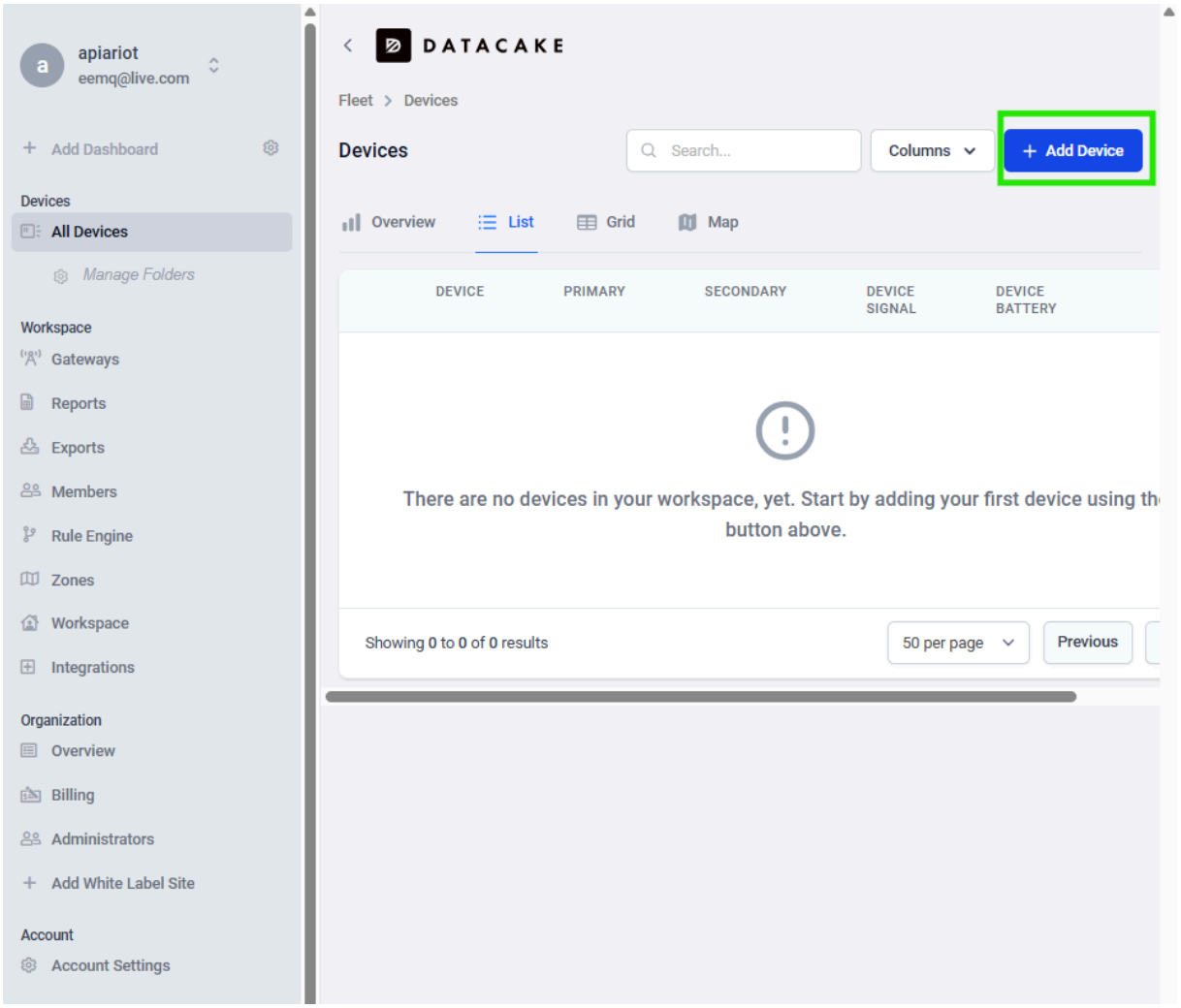
- Copia el token para TTN

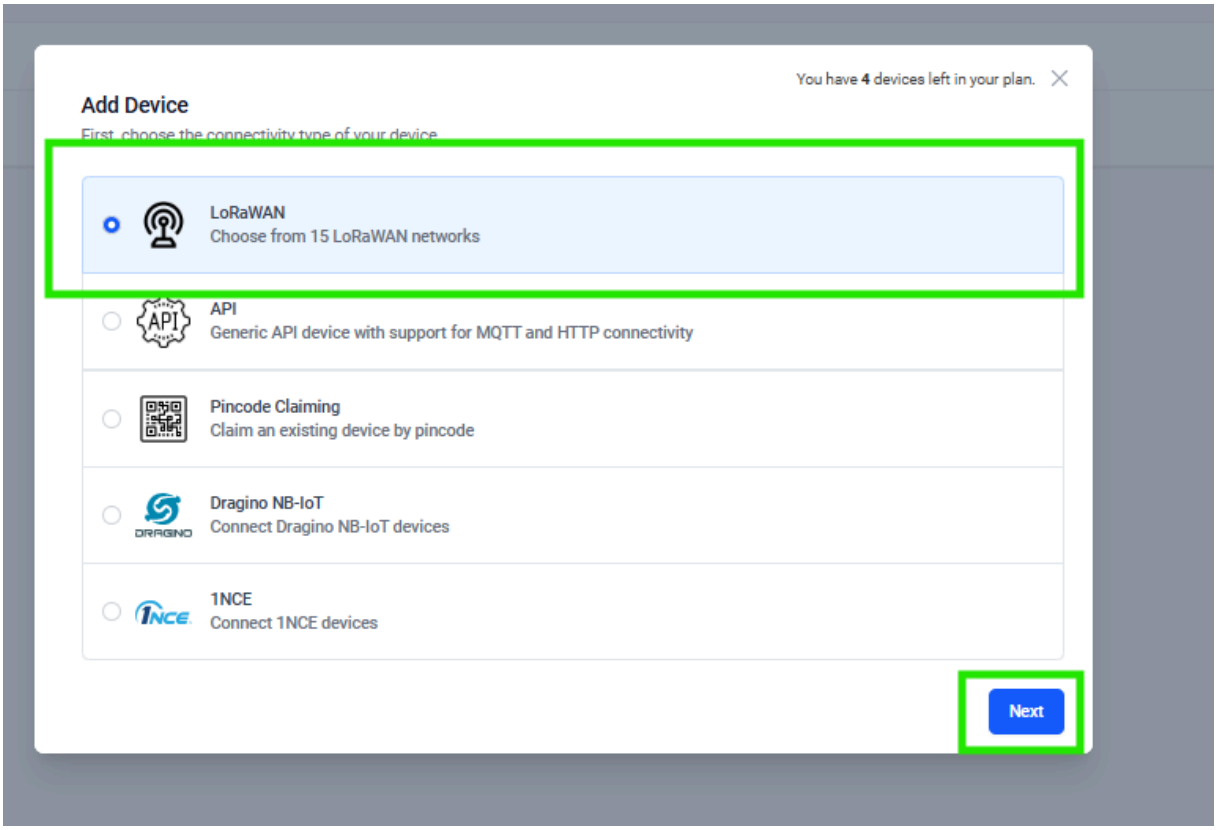


- Ya tienes el token para TTN.

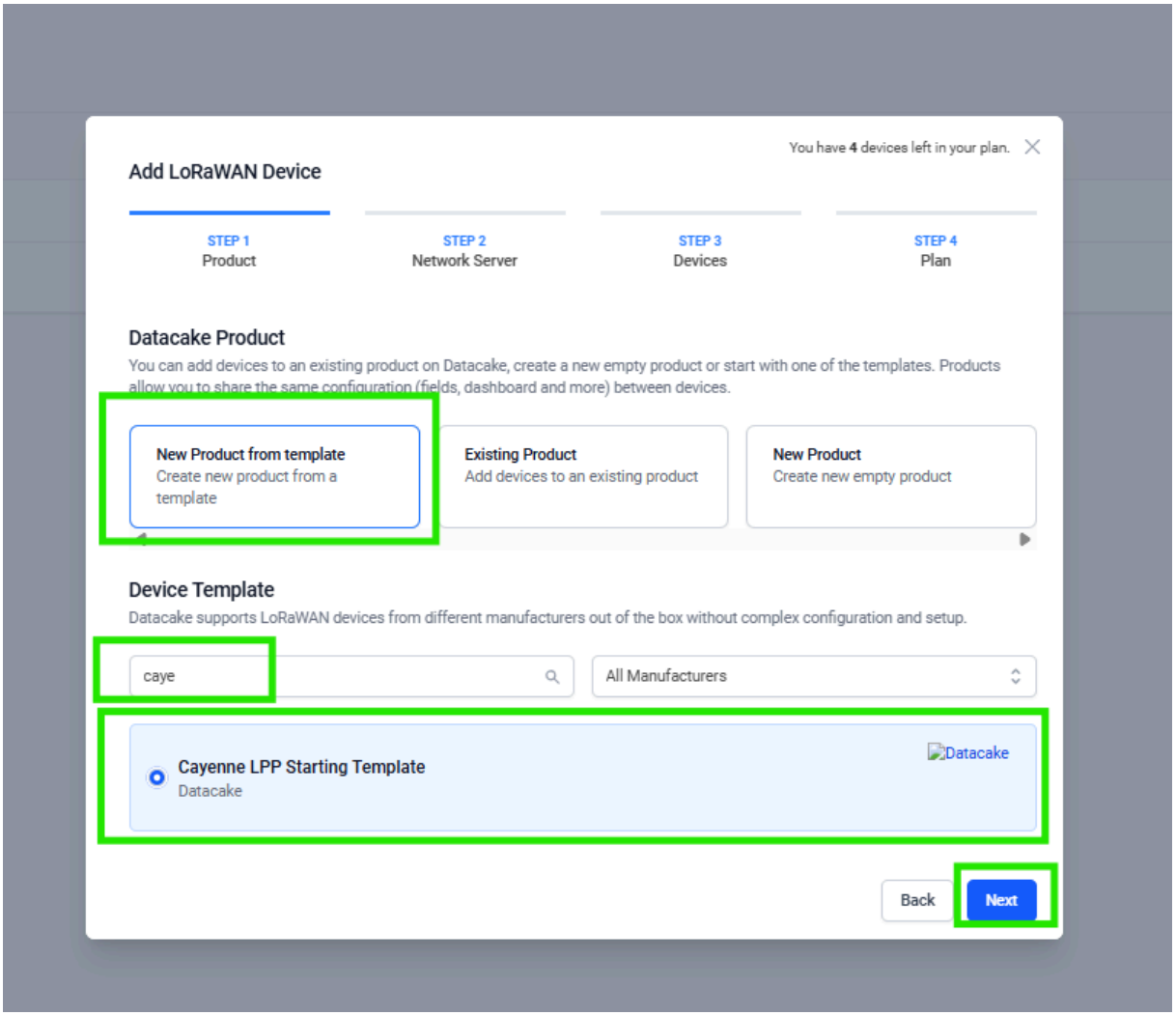


- Agrega tu dispositivo ahora en Datacake.




You have 4 devices left in your plan.
Add Device
First, choose the connectivity type of your device
LoRaWAN
Choose from 15 LoRaWAN networks
API
Generic API device with support for MQTT and HTTP connectivity
Pincode Claiming
Claim an existing device by pincode
Dragino NB-IoT
Connect Dragino NB-IoT devices
1NCE
Connect 1NCE devices
Next

- Utilice el servidor de red: eu1.cloud.thethings.network (Suiza)



SCHRITT 1
Produkt

SCHRITT 2
Netzwerk-Server

SCHRITT 3
Geräte

SCHRITT 4
Plan

Netzwerk-Server

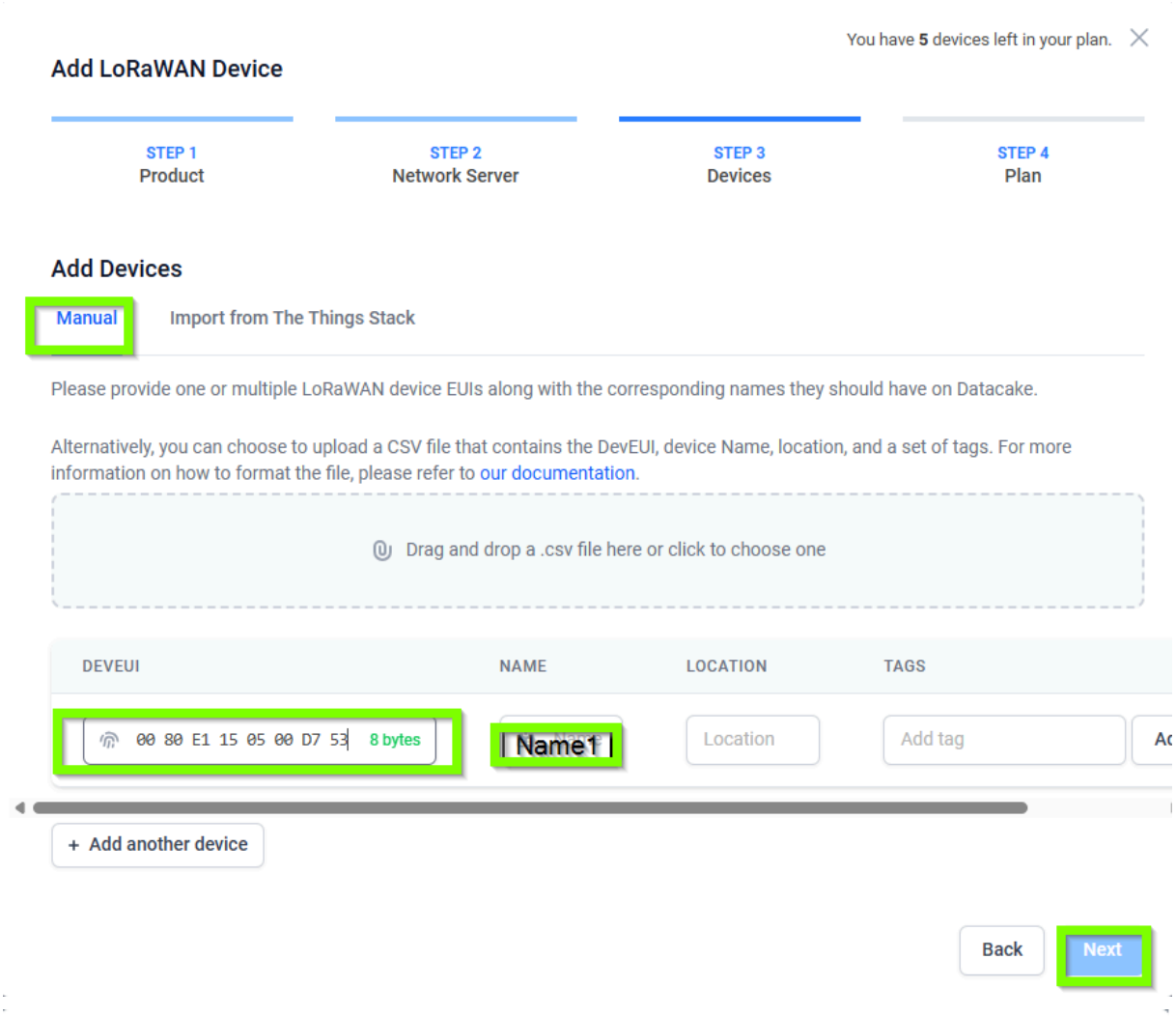
Bitte wählen Sie den LoRaWAN Network Server, mit dem Ihre Geräte verbunden sind.

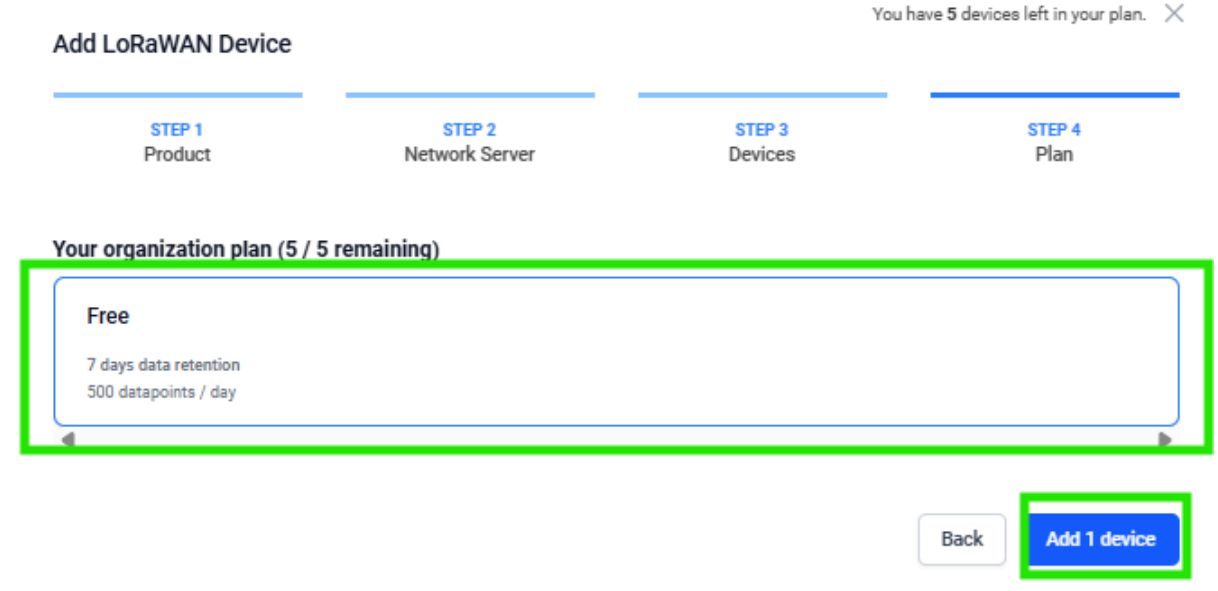
○	**Datacake LNS** AUTOMATIC SETUP Einfach starten und skalieren mit einem verwalteten LNS	Uplinks Downlinks
◉	**The Things Stack V3** TTN V3 / Things Industries	Uplinks Downlinks
○	**ThingPark**	Uplinks Downlinks
○	**KPN**	Uplinks Downlinks
○	**LORIOT**	Uplinks Downlinks
○	**ChirpStack**	Uplinks Downlinks
○	**Helium** Verwenden Sie Ihre eigene Konsole	Uplinks Downlinks

Anzeigen von **1** bis **6** von **14** Ergebnissen

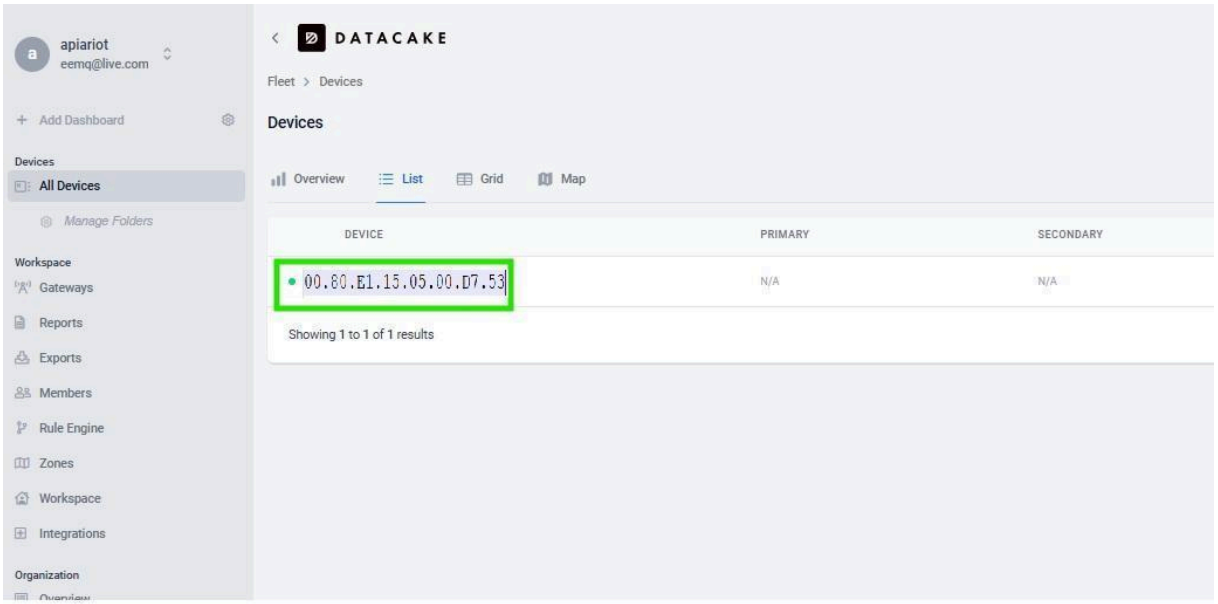
Vorherige Weiter

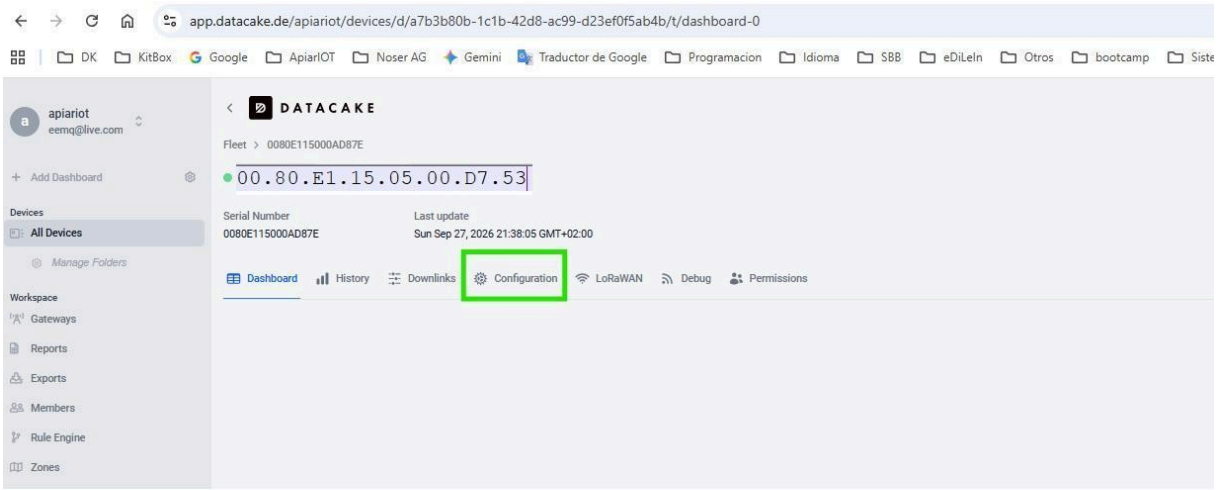
Zurück Weiter


You have 5 devices left in your plan.
Add LoRaWAN Device
STEP 1
Product
STEP 2
Network Server
STEP 3
Devices
STEP 4
Plan
Add Devices
Manual
Import from The Things Stack
Please provide one or multiple LoRaWAN device EUIs along with the corresponding names they should have on Datacake.
Alternatively, you can choose to upload a CSV file that contains the DevEUI, device Name, location, and a set of tags. For more information on how to format the file, please refer to our documentation.
Drag and drop a .csv file here or click to choose one
DEVEUI
NAME
LOCATION
TAGS
00 80 E1 15 05 00 D7 53
8 bytes
Name1
Location
Add tag
+ Add another device
Back
Next

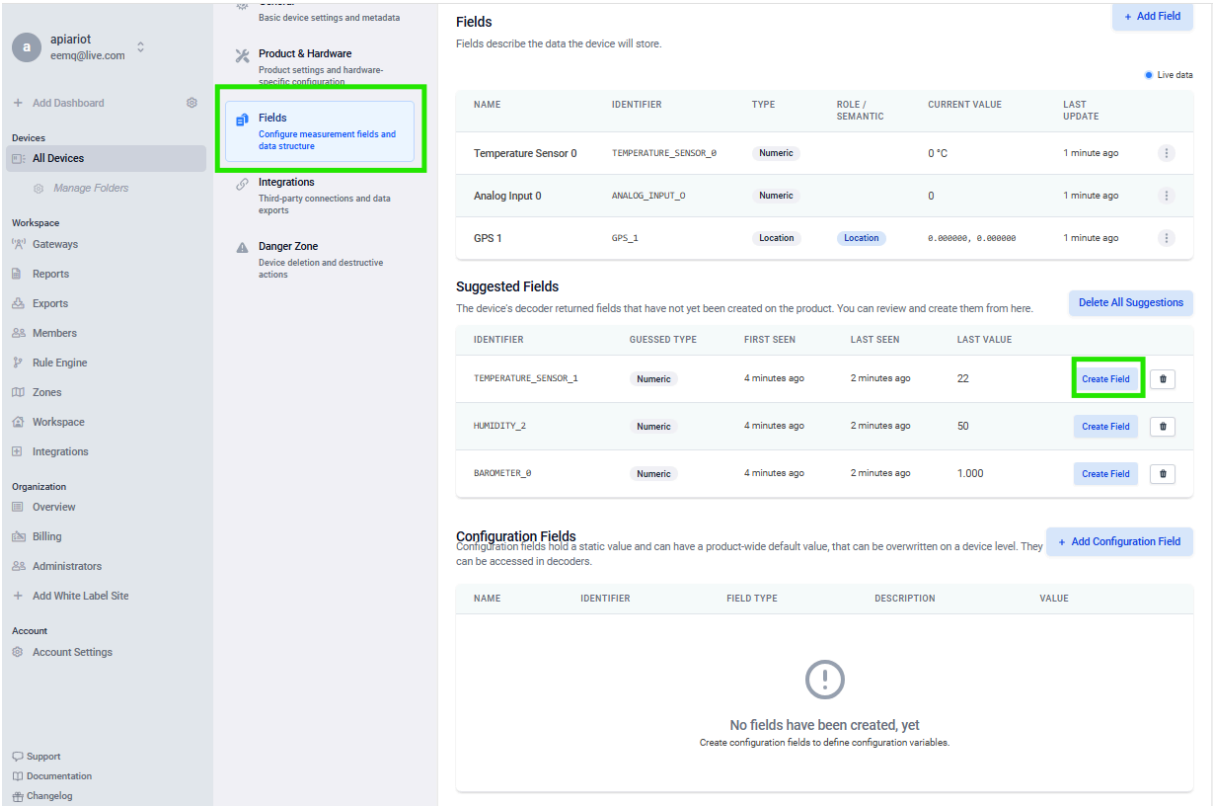

You have 5 devices left in your plan.
Add LoRaWAN Device
STEP 1
Product
STEP 2
Network Server
STEP 3
Devices
STEP 4
Plan
Your organization plan (5 / 5 remaining)
Free
7 days data retention
500 datapoints / day
Back
Add 1 device

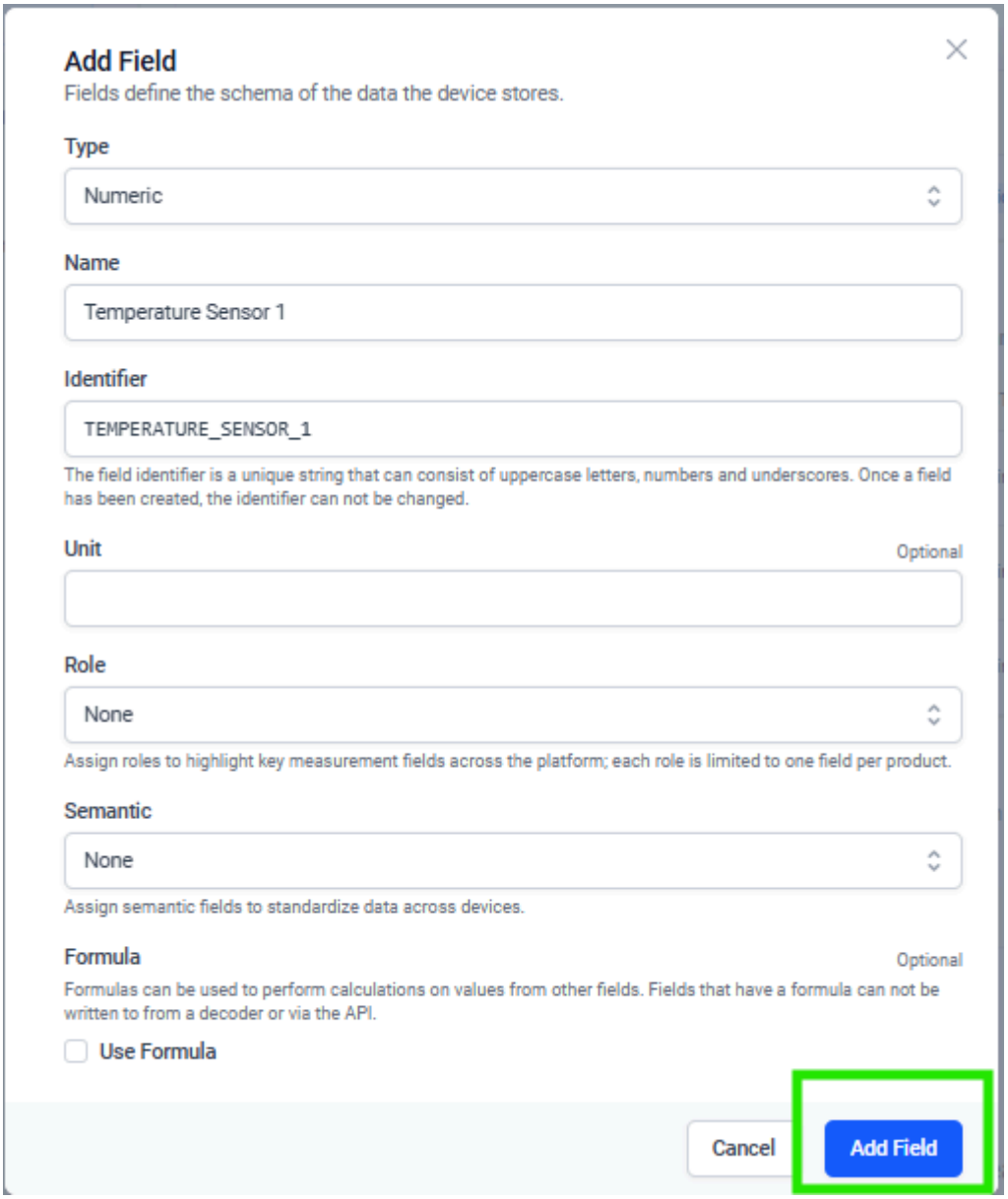
- Vuelve a activar tu "End Node".



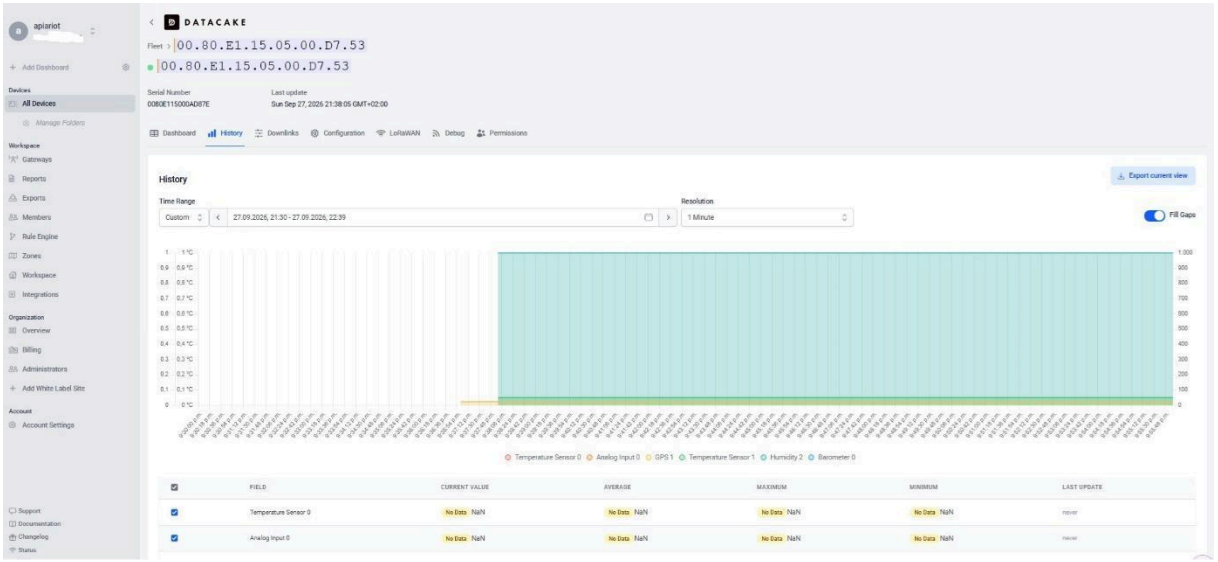


- Espere hasta que se reciban los datos.





- Ahora debería poder ver los datos de su placa de desarrollo en Datacake.



- Y también en TTN

